先進封裝創新推動AI 發展

Yin Chang
ASE, Inc.

December 02, 2025

AI 時代已經到來



HEALTHCARE



TELECOMMUNICATIONS



RETAIL

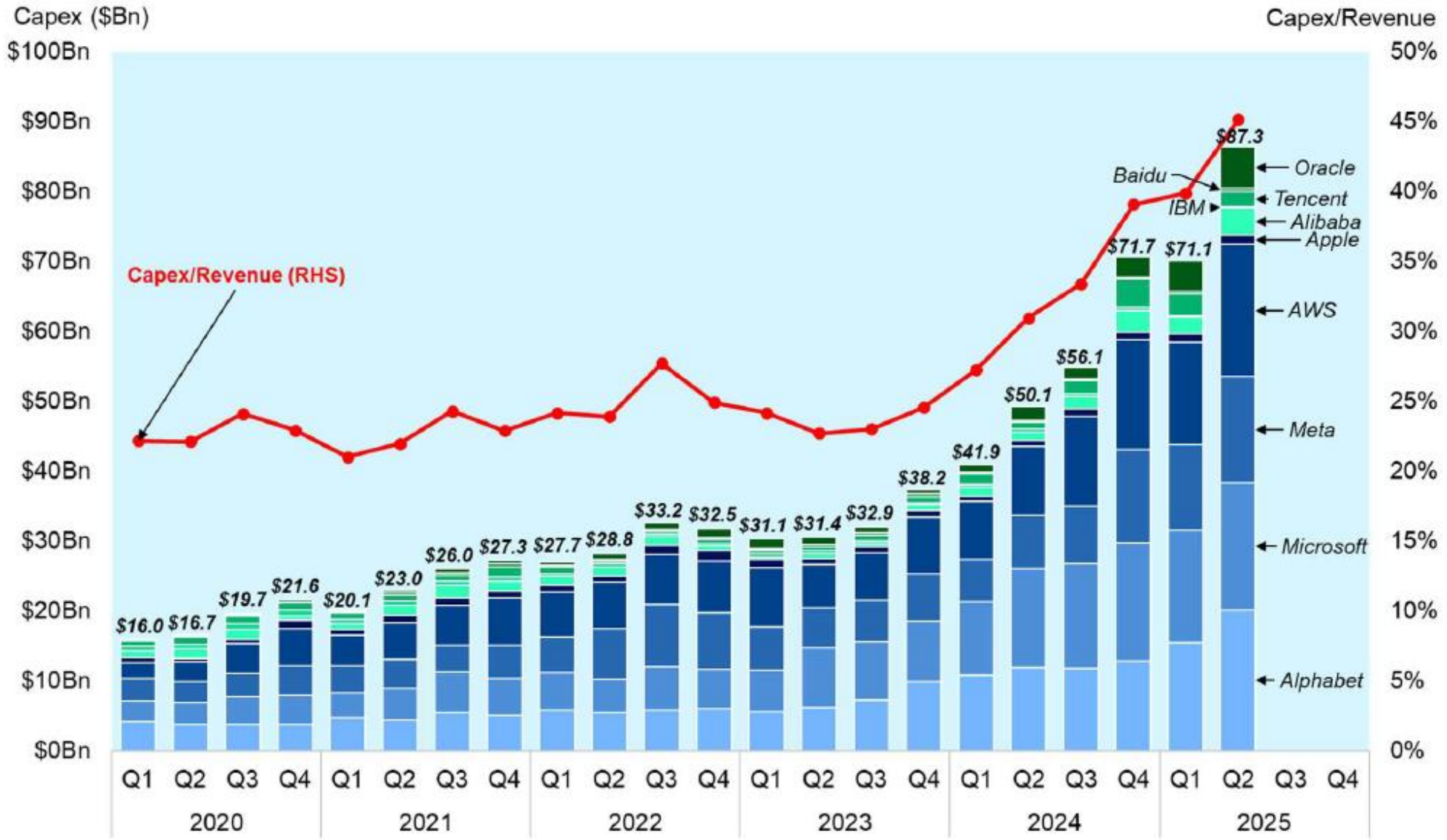


FINANCIAL SERVICES

- AI applications permeating global life, creating new efficiencies and new markets, from data center to edge devices.
- AI economy projected to soar from $189 billion in 2023 to $4.8 trillion by 2033 – a 25-fold increase in just a decade
- AI adoption could boost global GDP by 15% by 2035, based on game-changing capabilities and safe deployment.
- Unprecedented data generation is driving global datasphere trajectory towards 200ZB by 2030

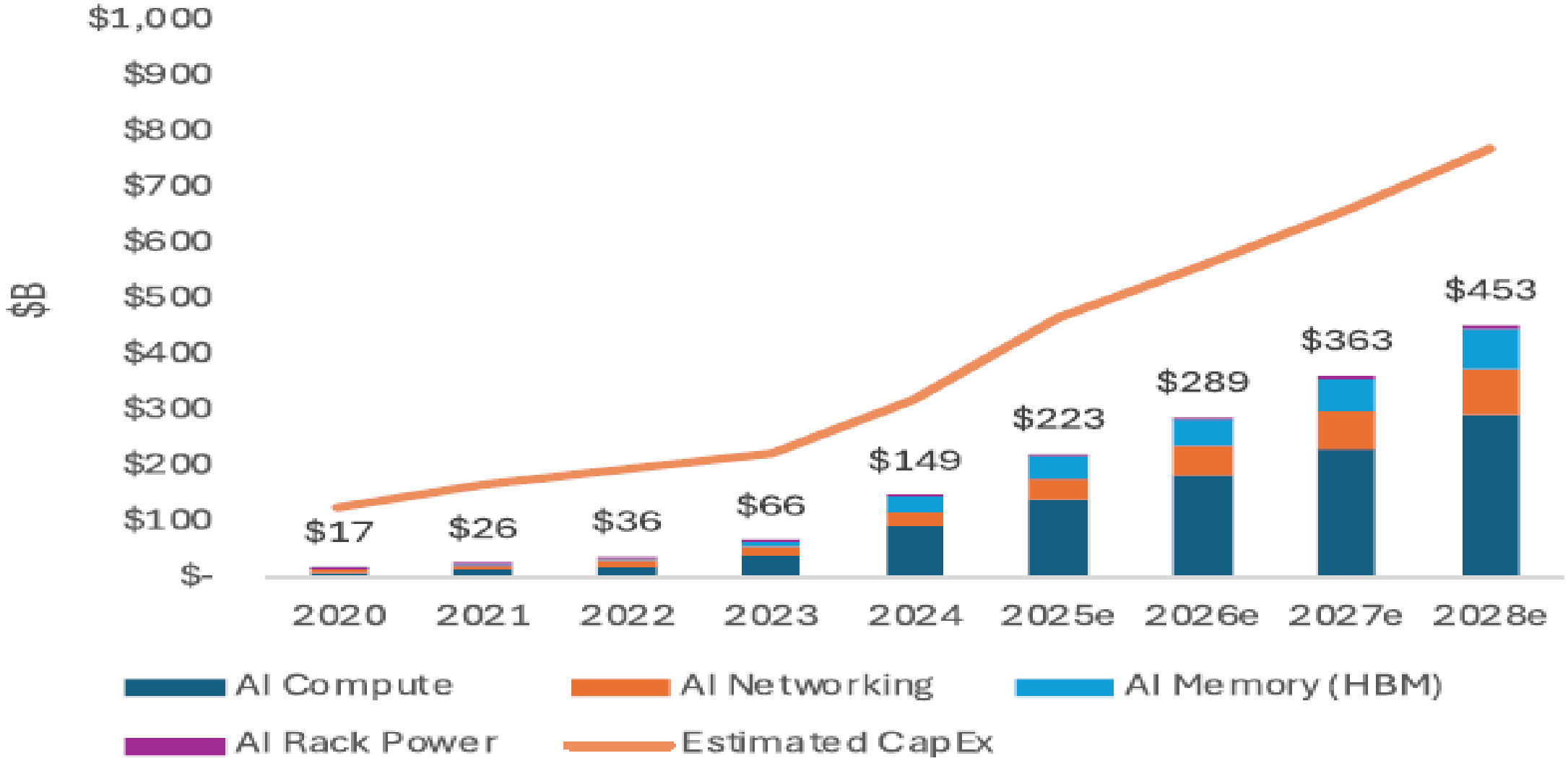
Source:
UN Trade and Development (UNCTAD), PwC, IDC 2025

AI 資本支出爆炸性增長



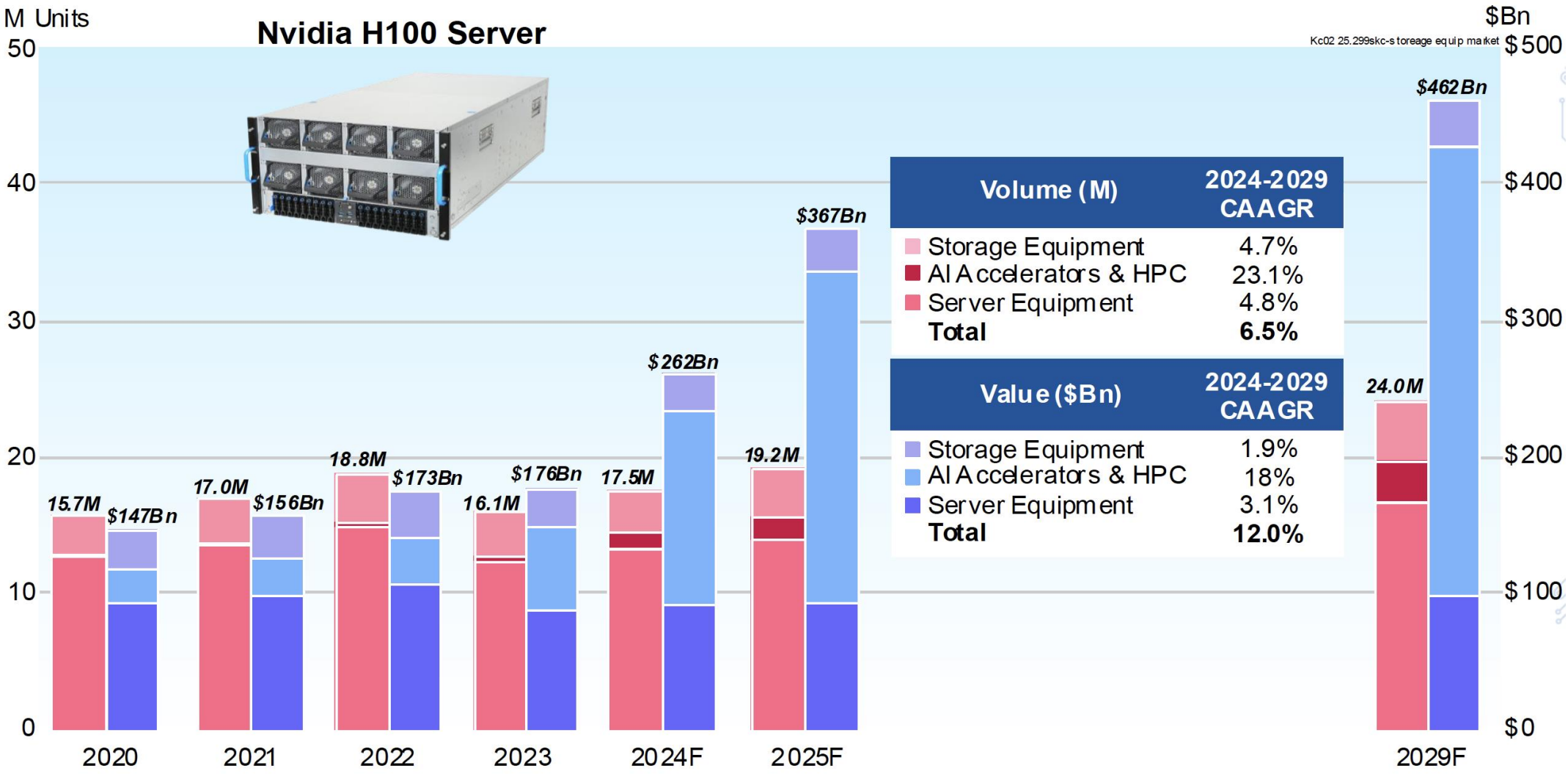
Source: Prismark

資料中心資本支出與AI 半導體



Source :IDC

AI 設備市場


M Units
$Bn
Nvidia H100 Server
Kc02 25.299skc-storeage equip market
50
40
30
20
10
0
$500
$400
$300
$200
$100
$0
15.7M
$147Bn
17.0M
$156Bn
18.8M
$173Bn
16.1M
$176Bn
17.5M
$262Bn
19.2M
$367Bn
24.0M
$462Bn
2020
2021
2022
2023
2024F
2025F
2029F
Volume (M)
2024-2029 CAAGR
Storage Equipment 4.7%
AI Accelerators & HPC 23.1%
Server Equipment 4.8%
Total 6.5%
Value ($Bn)
2024-2029 CAAGR
Storage Equipment 1.9%
AI Accelerators & HPC 18%
Server Equipment 3.1%
Total 12.0%

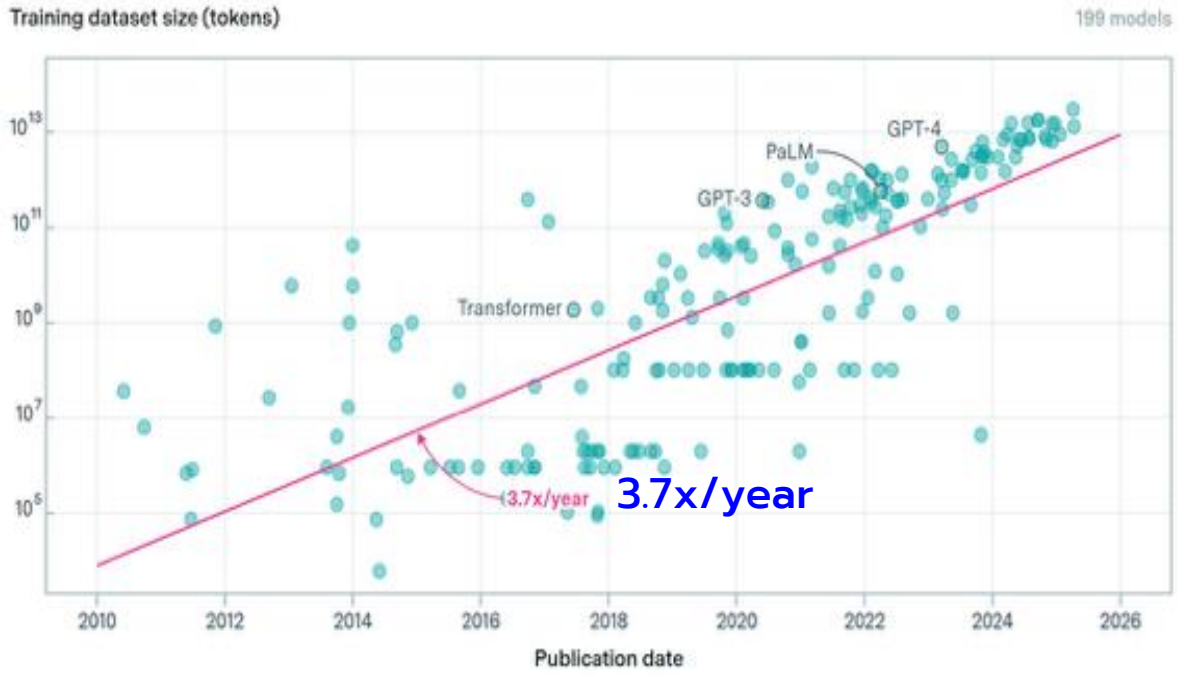
AI 時代增長的需求與挑戰

Performance	Memory	Area	Power	Thermal
2x - 7x increased compute workload	**2x – 10x** network bandwidth & memory capacity	**2x – 12x** reticle **2x – 6x** pkg size (120x180 mm^2)	Exascale **21 MW** to zettascale **500 MW** power delivery	**~3kW** thermal cooling

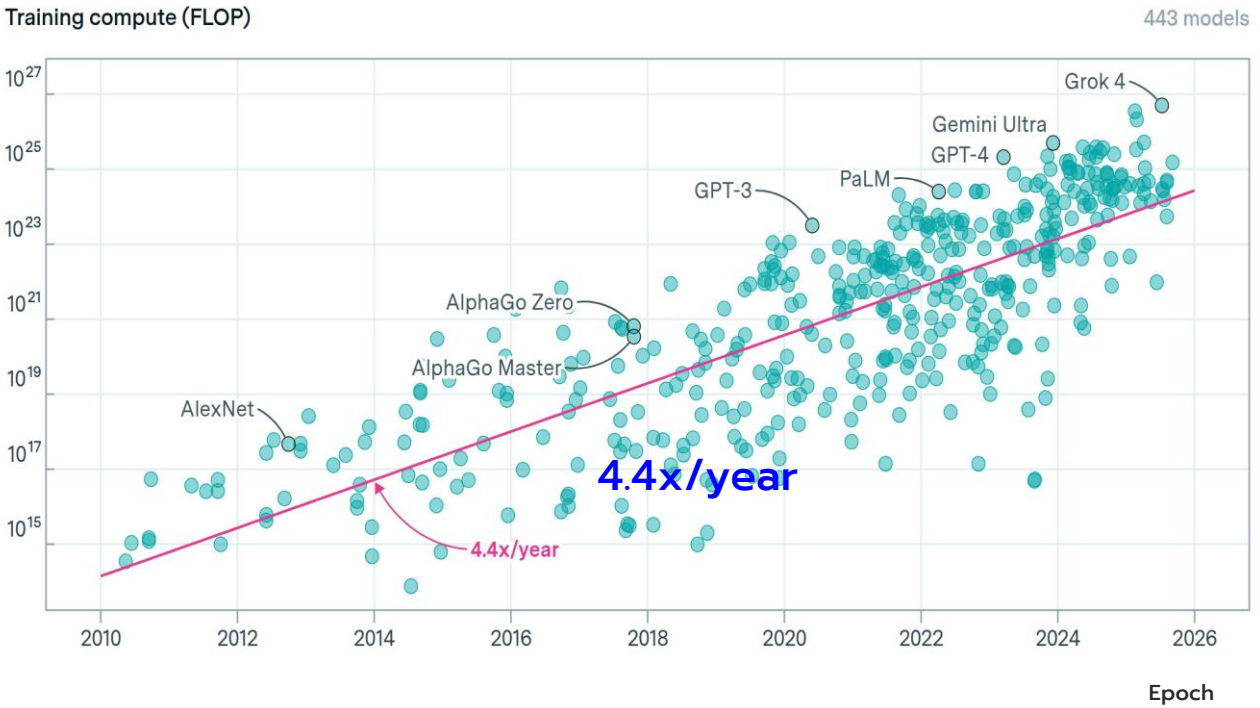
AI 產業推動高效能需求

- Across all domains of AI, dataset size for train models doubles every 6 months with tens of trillions of words
- High-performance compute to train models grew 4-5x yearly 2010~2026 with 1.7x/year increase chip quantity & 1.4x/year improvement in performance per chip

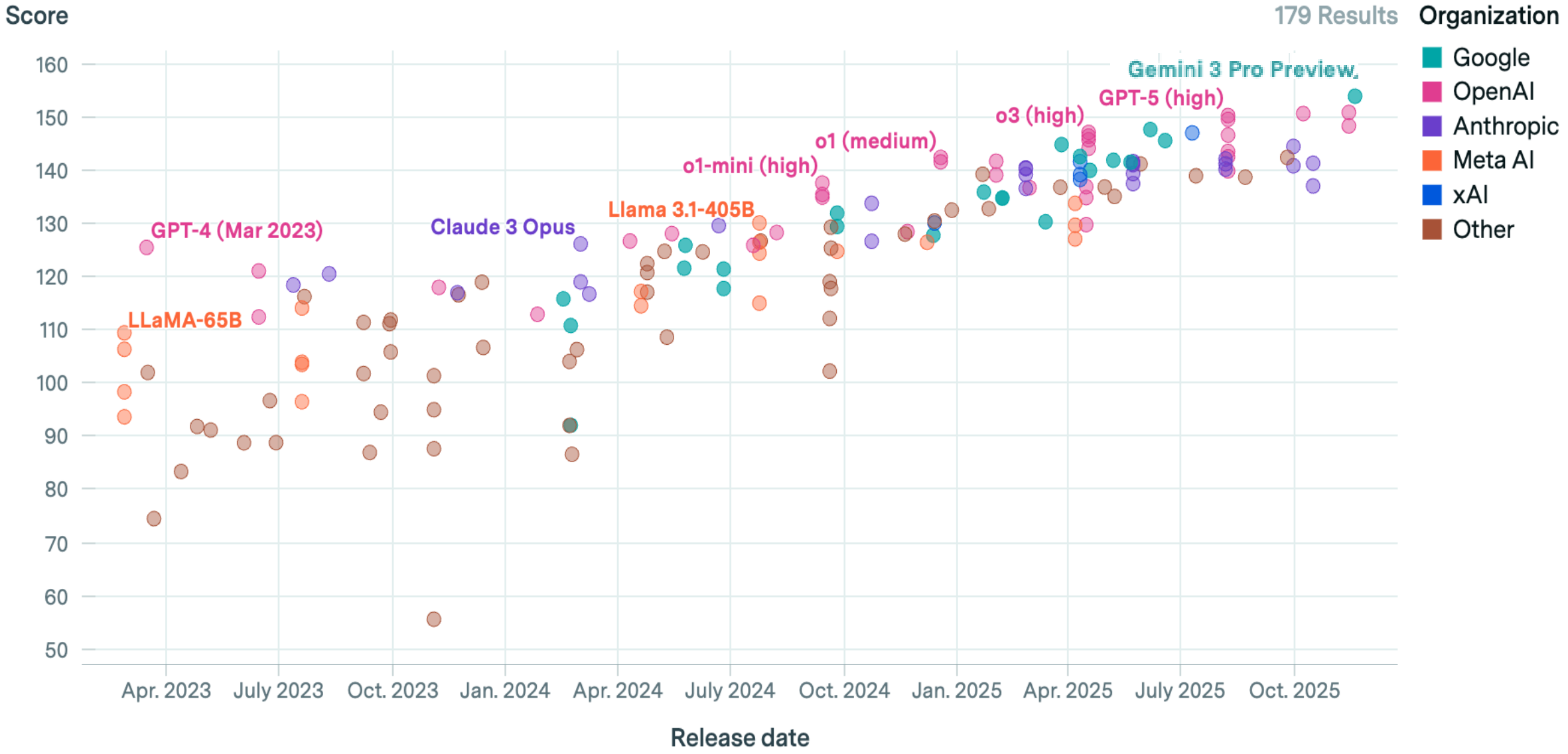
Size of Dataset



Super Compute Performance



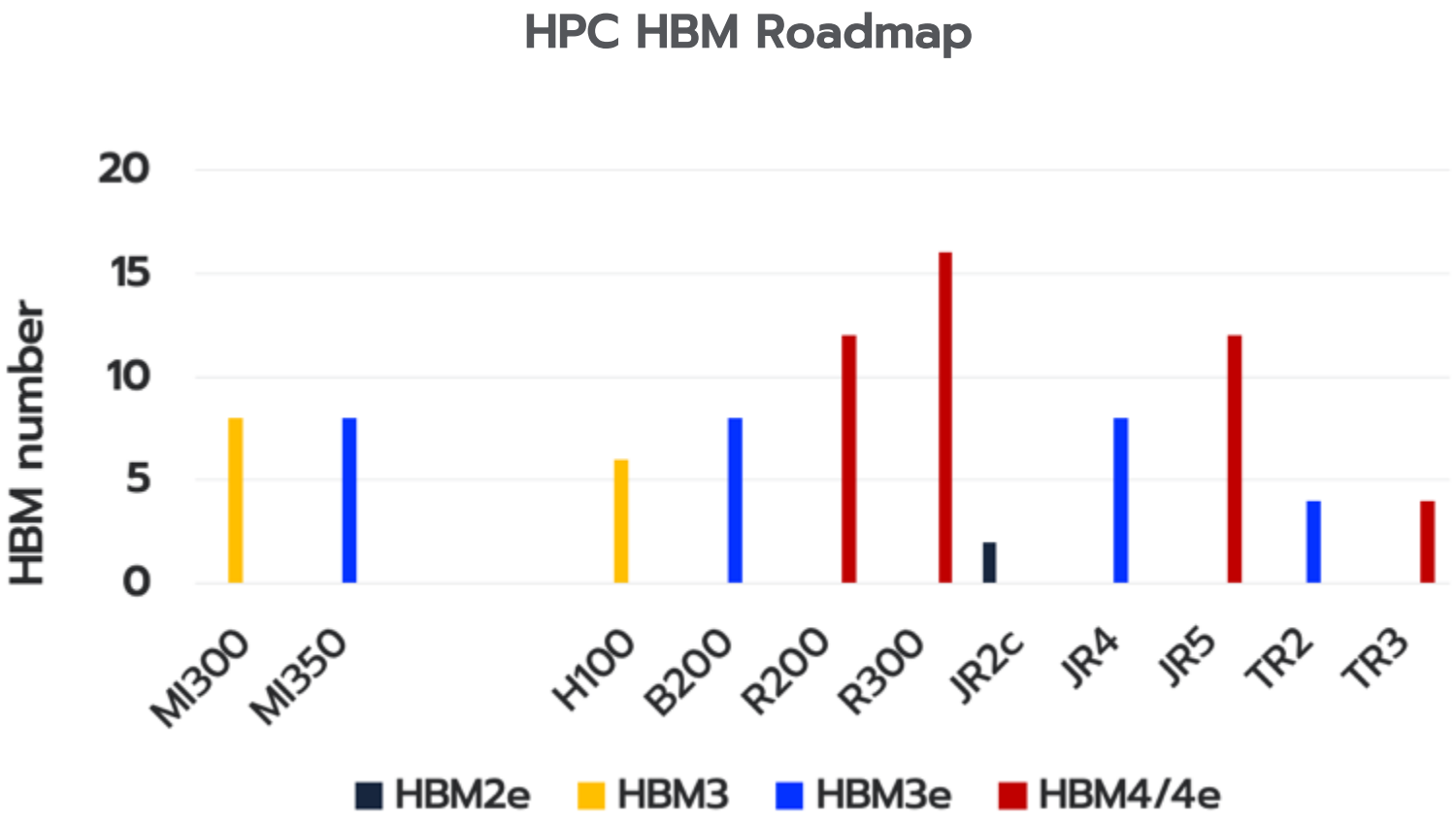
AI 龍頭模型效能比較圖

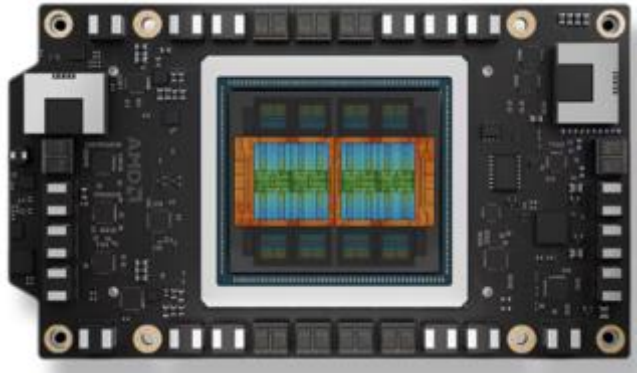


Source Epoch AI

HPC HBM 技術路線圖

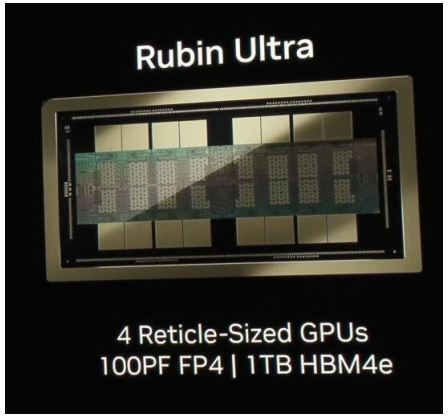
- HBM3e to HBM4/4e will increase from 8 to 12 to 16 by 2027~2028
- Larger module size required for greater HBM integration





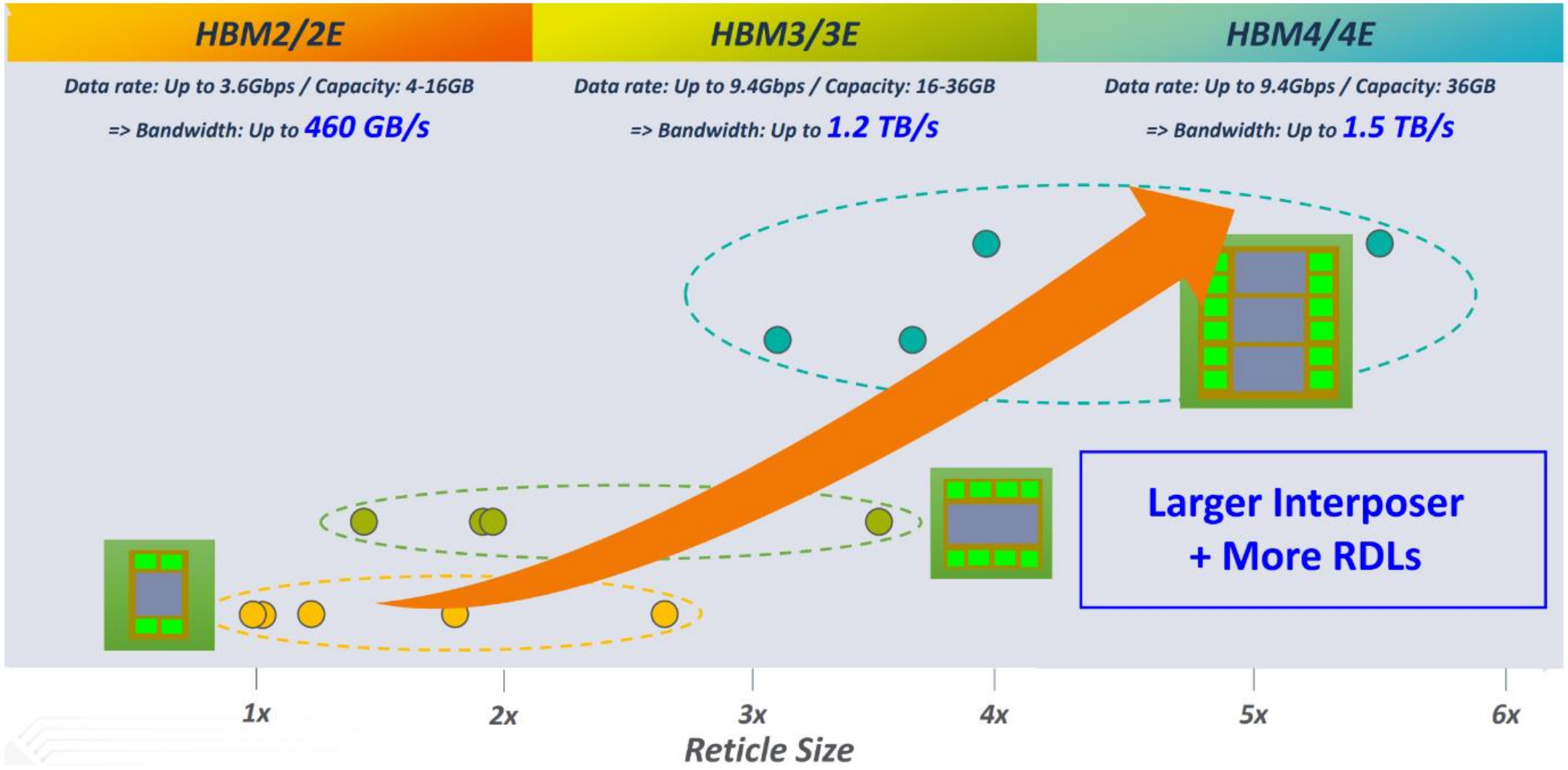
AMD MI350 (N3/N6, 18 chiplets, SOIC + CoWoS-S; 8HBM3e 288GB)

Source AMD



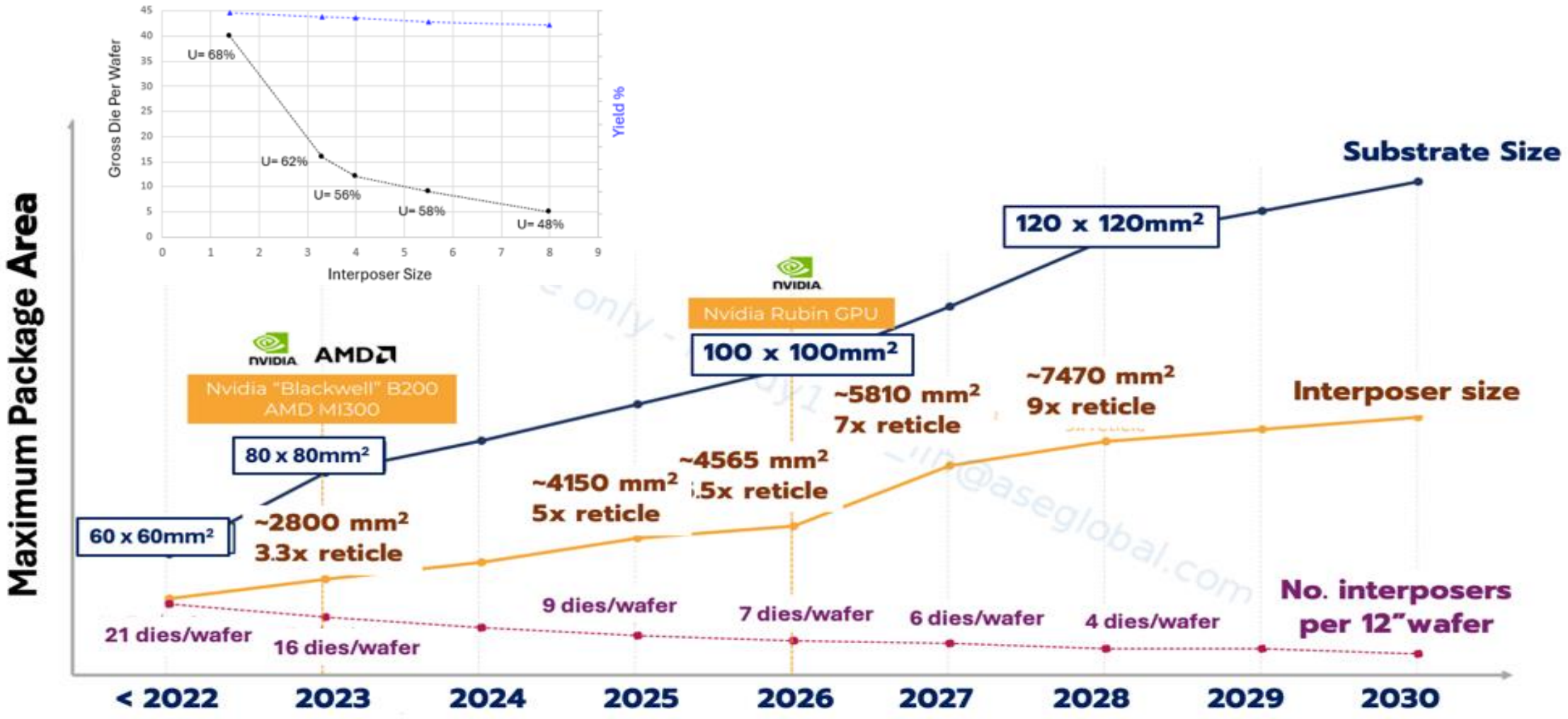
Nvidia Rubin Ultra CoWoS-L 16HBM4E,
Interposer size 124x50mm
Substrate size 153x77.5mm Source Nvidia

HPC & HBM 整合趨勢


HBM2/2E
HBM3/3E
HBM4/4E
Data rate: Up to 3.6Gbps / Capacity: 4-16GB
=> Bandwidth: Up to 460 GB/s
Data rate: Up to 9.4Gbps / Capacity: 16-36GB
=> Bandwidth: Up to 1.2 TB/s
Data rate: Up to 9.4Gbps / Capacity: 36GB
=> Bandwidth: Up to 1.5 TB/s
Larger Interposer
+ More RDLs
1x
2x
3x
4x
5x
6x
Reticle Size

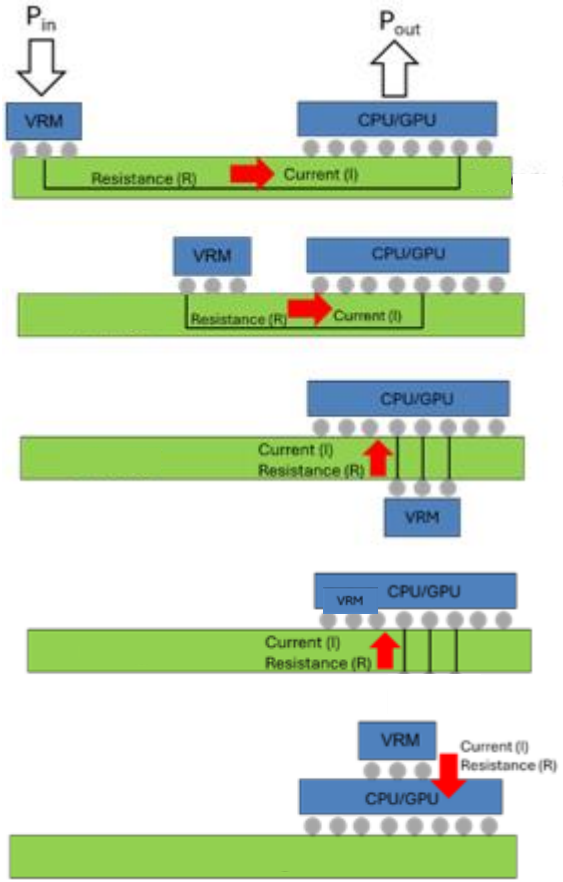
大型模組面臨的主要挑戰

- Yield and utilization decreases when module size increases



Yole

電力傳輸挑戰



MBVR – Lateral PD

MBVR – Vertical PD

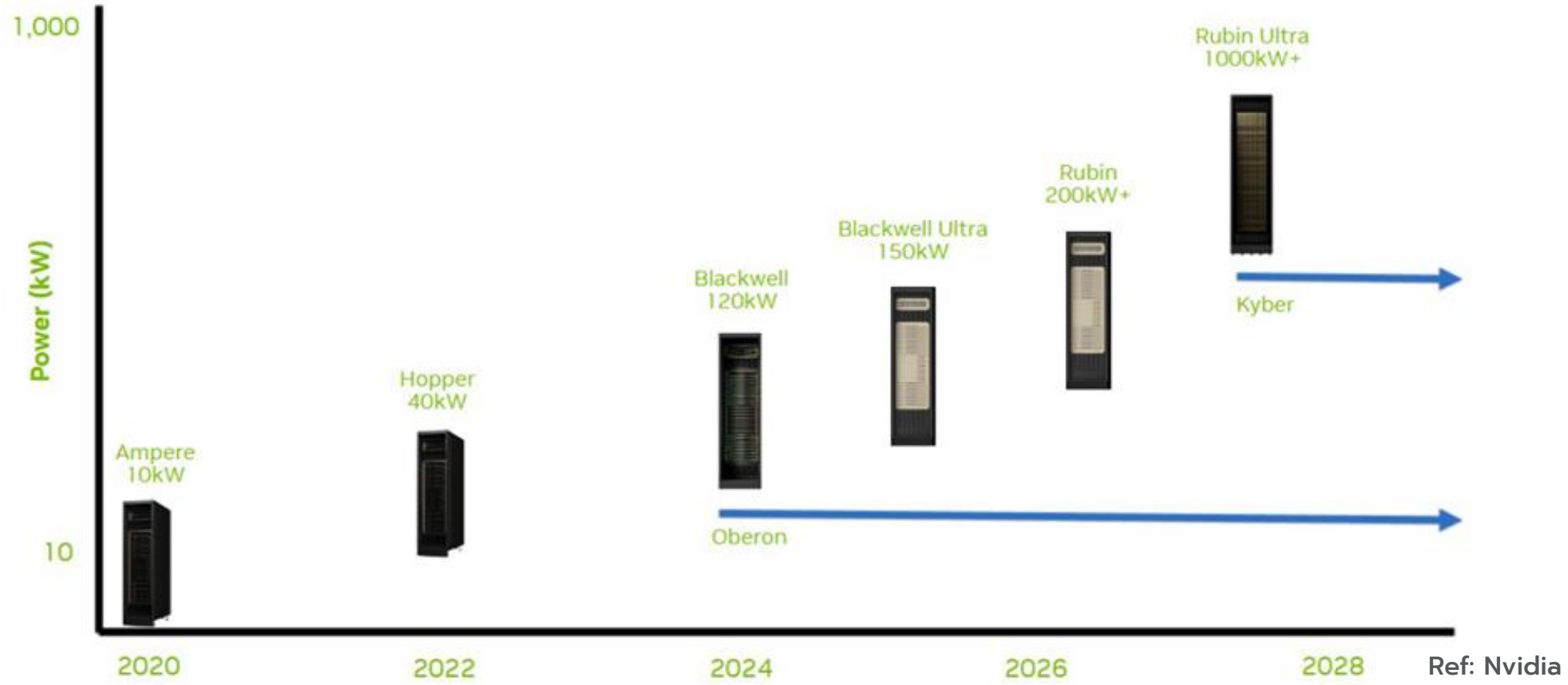
IVR Options
- Substrate, RDL
- Low, High Voltage

Backside PD

- PCB, Package Resistance → Routing losses
  - Distance of VRM to Load, voltage levels
  - Doubling of current, results in 4X power loss (I^2R)
  - Package power can be >10% of total TDP, with large IR drops
- Lower operating voltage and increasing voltage domains → higher load currents
- PDN impedance (Z) over frequency
  - Z needs to decrease in proportion to maintain acceptable voltage droop
  - Under package capacitor banks seeing diminishing returns in lowering Z to match current step/speed increases
- High phase count external VRM is spatially inefficient

MBVR – Motherboard Voltage Regulator; PD – Power Delivery; IVR – Integrated Voltage Regulator; RDL – Redistribution Layer

AI 運算機櫃電源路線圖

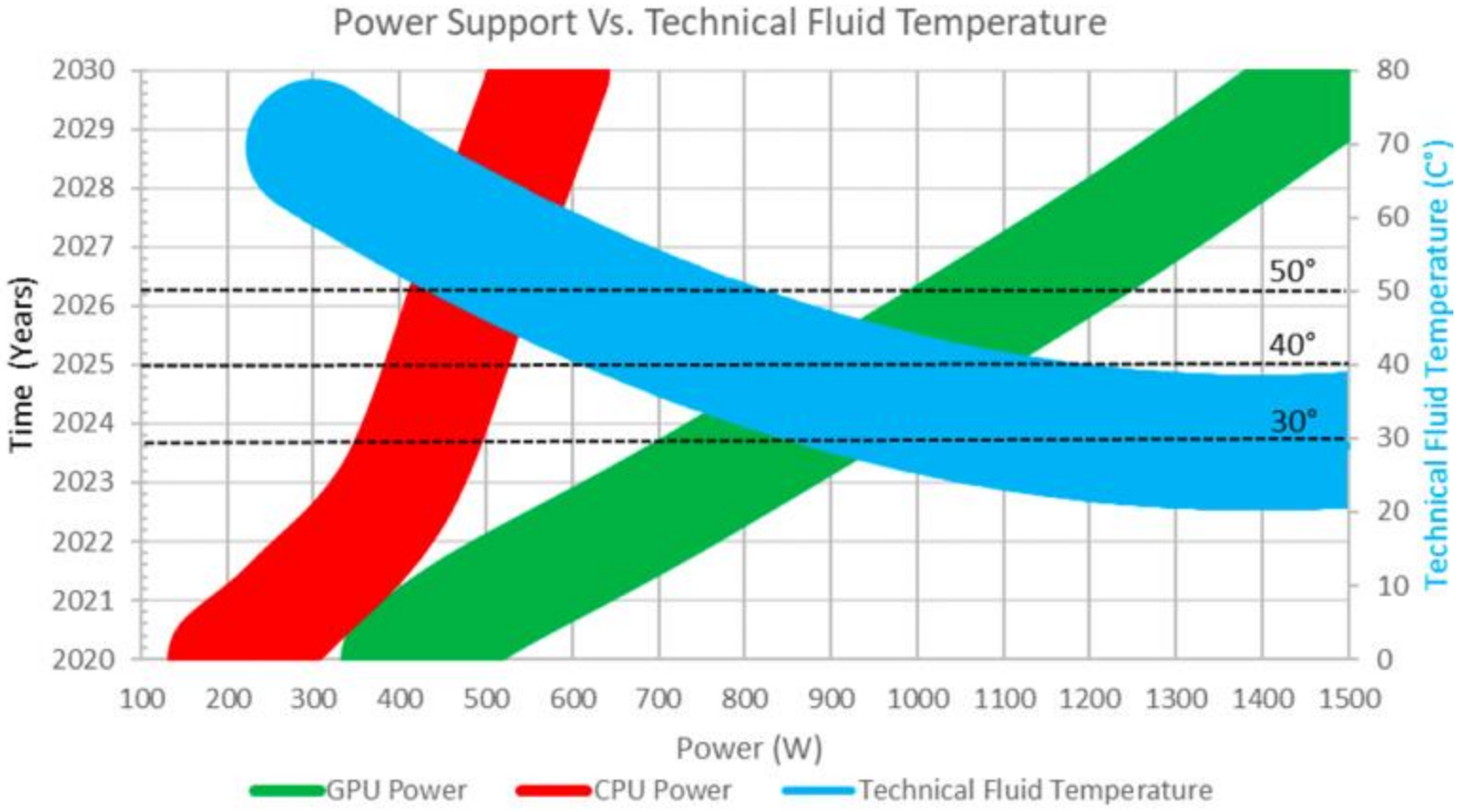


Ref: Nvidia

- AI workloads driving rack wattage from KW to MW
- Fundamental shift in power infrastructure to support the scale and power demands

AI 的散熱需求

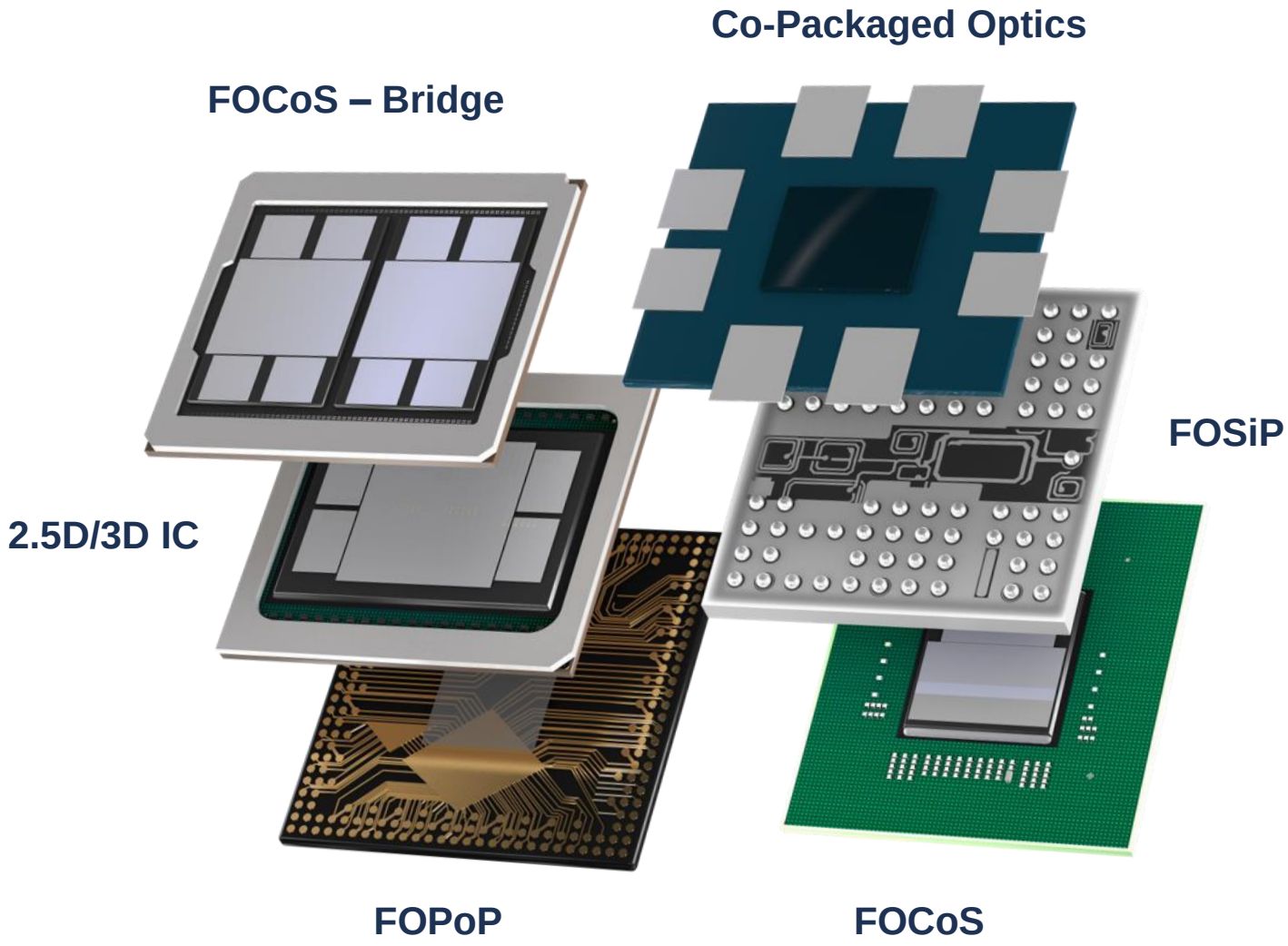
- CPU & GPU power demands rapidly increasing with GPU power expected > 1500W & CPU >600W by 2028
- Technical fluid temperature requires lower temp for effective heat dissipation while power increases
- Thermal management becomes the bottleneck for more efficient liquid cooling or lower temperature operations



Source: AMD

VIPack™解決方案:

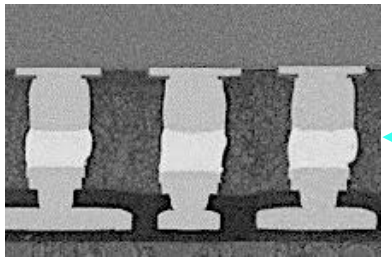
an advanced packaging platform designed to enable vertically integrated package solutions, featuring six technology pillars and underpinned with ASE's Integrated Design Ecosystem™ (IDE)



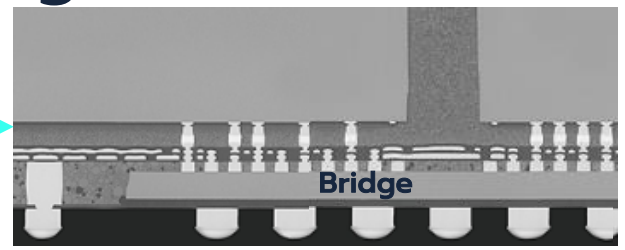
FOCoS and FOCoS-Bridge

FOCoS

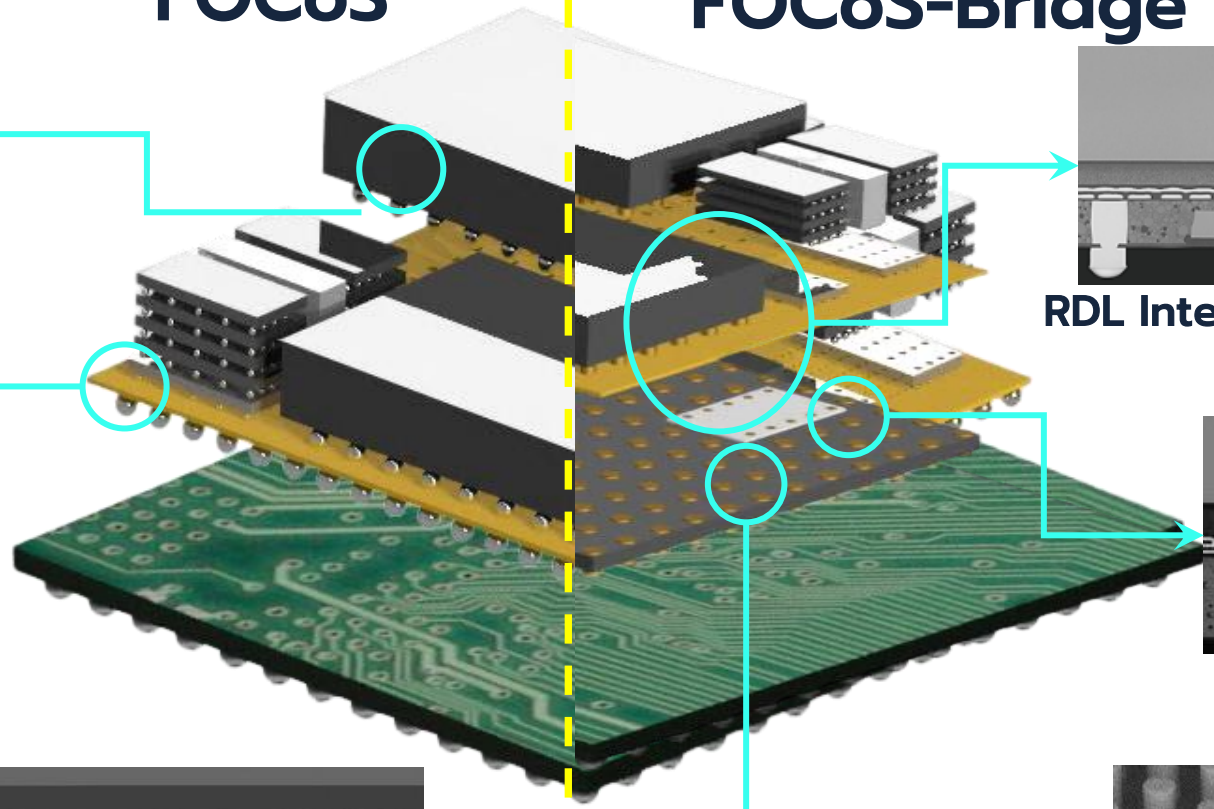
FOCoS-Bridge

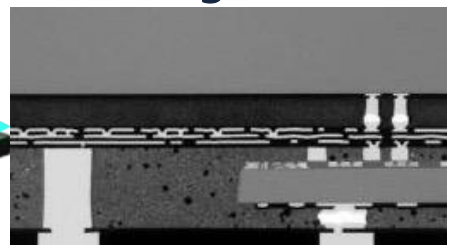


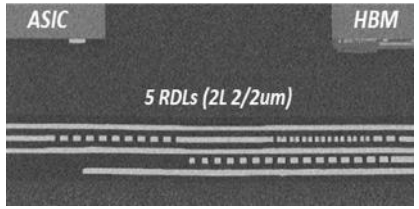
CoW uBump

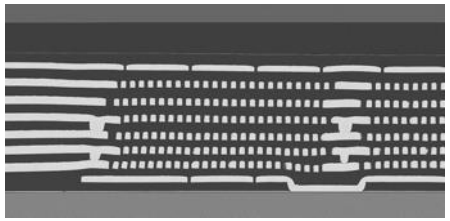


RDL Interposer with Embedded Bridge



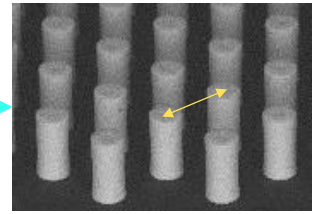




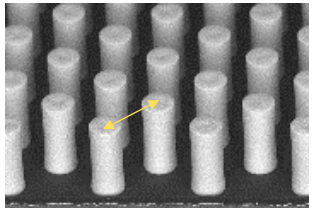


RDL Interposer

C4 bump pitch



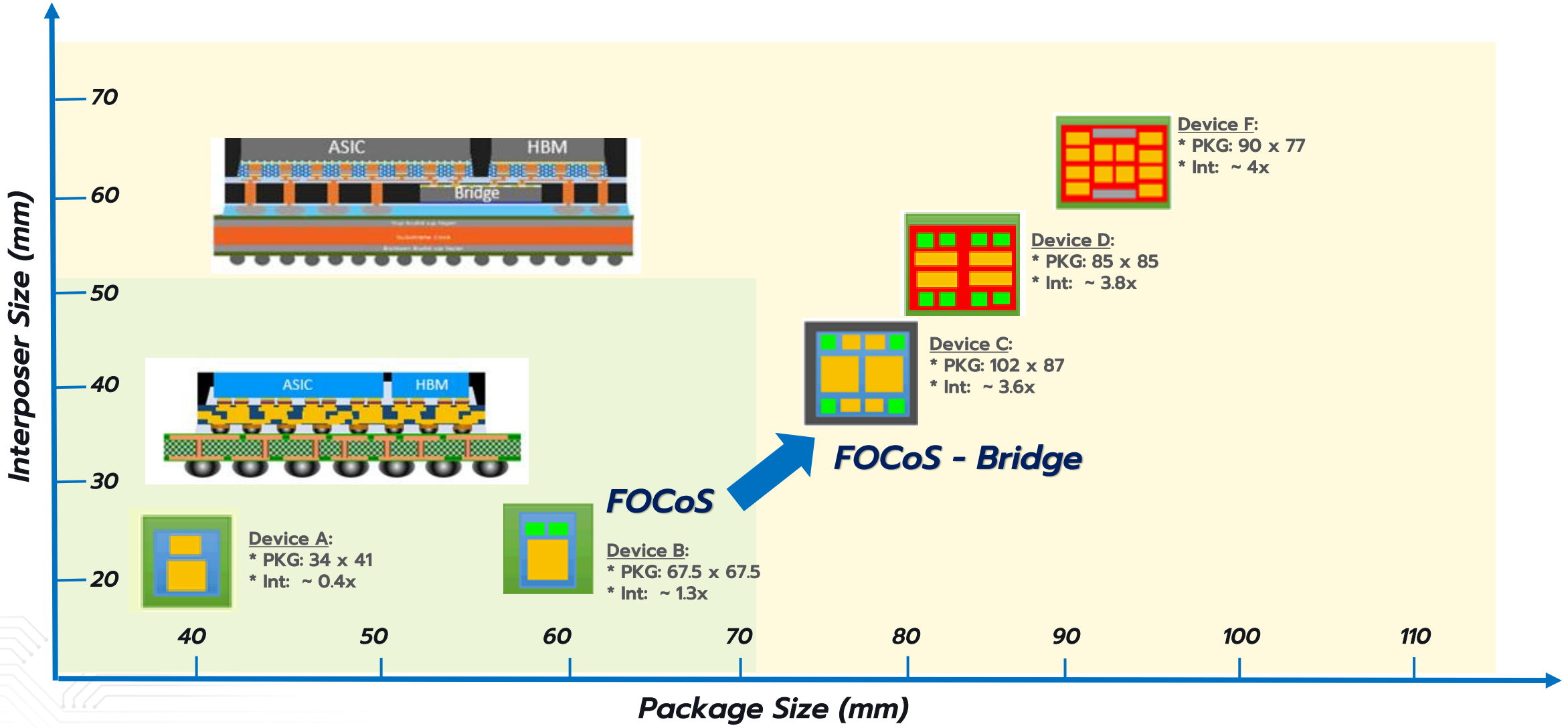
150um



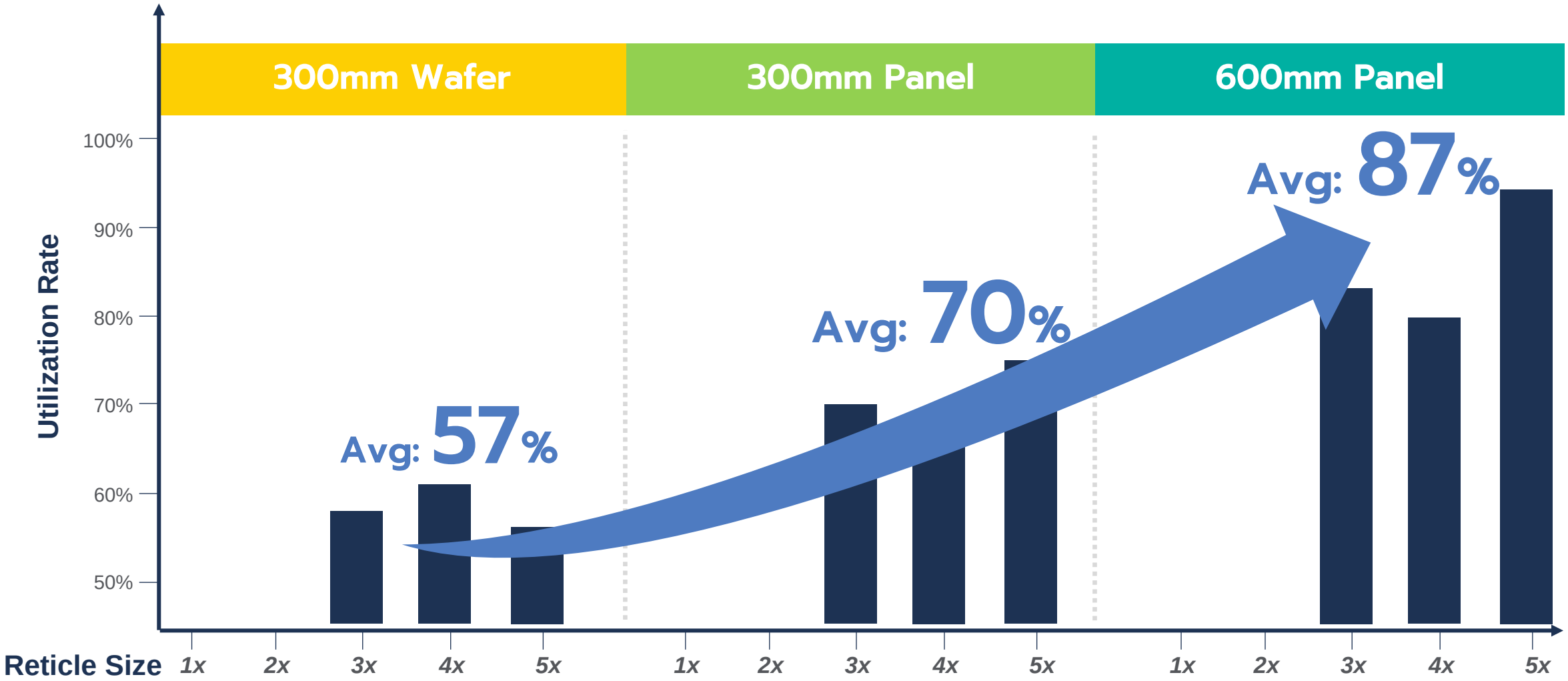
130um

FOCoS-B: FOCoS 衍生平台

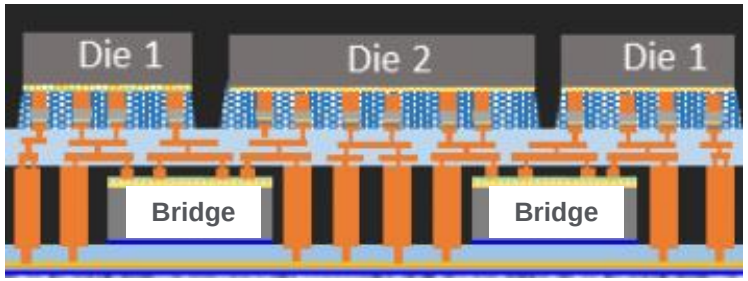
- Si bridge die provides high density D2D interconnect (< 1 um L/S) & enables embedded components – DTC, Power, Active, Passives for power delivery efficiency and performance improvement
- FOCoS-B provide larger scaling integration for larger reticle size (>3.3x) and larger package size (>70~150)

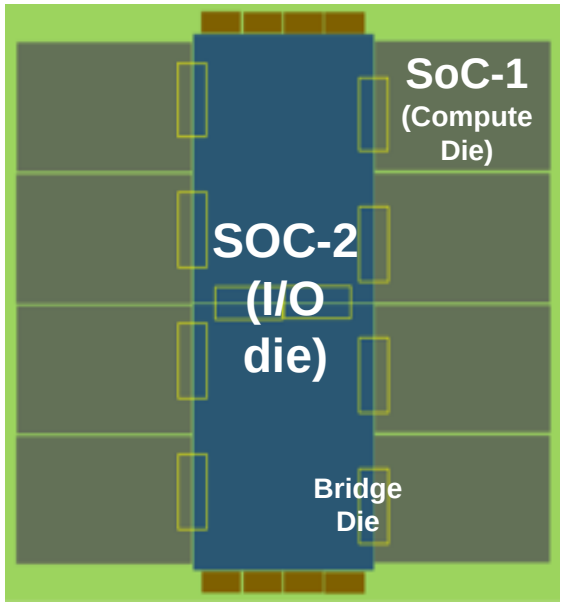


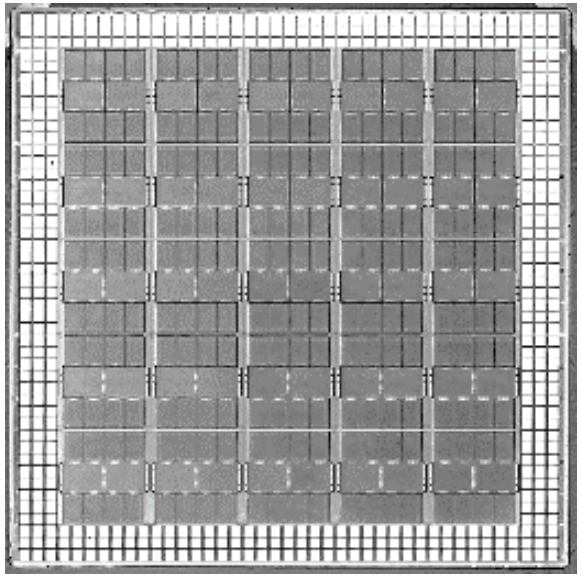
面板利用率與光罩尺寸


300mm Wafer
300mm Panel
600mm Panel
100%
90%
80%
70%
60%
50%
Utilization Rate
Avg: 57%
Avg: 70%
Avg: 87%
Reticle Size
1x
2x
3x
4x
5x
1x
2x
3x
4x
5x
1x
2x
3x
4x
5x

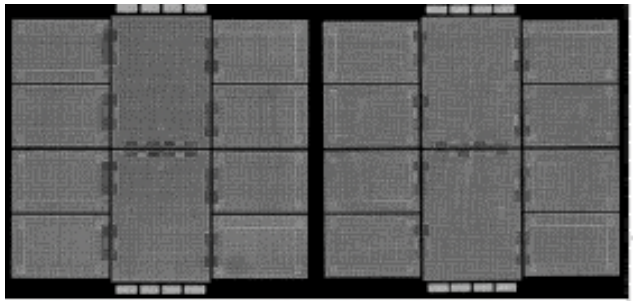
Panel FOCoS-Bridge



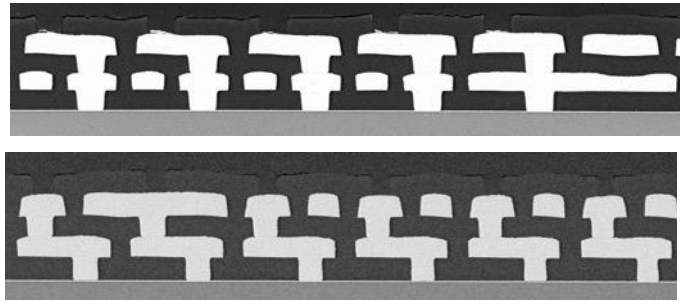




ubump pitch: 45um;
C4 bump pitch 150um

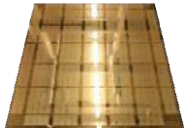


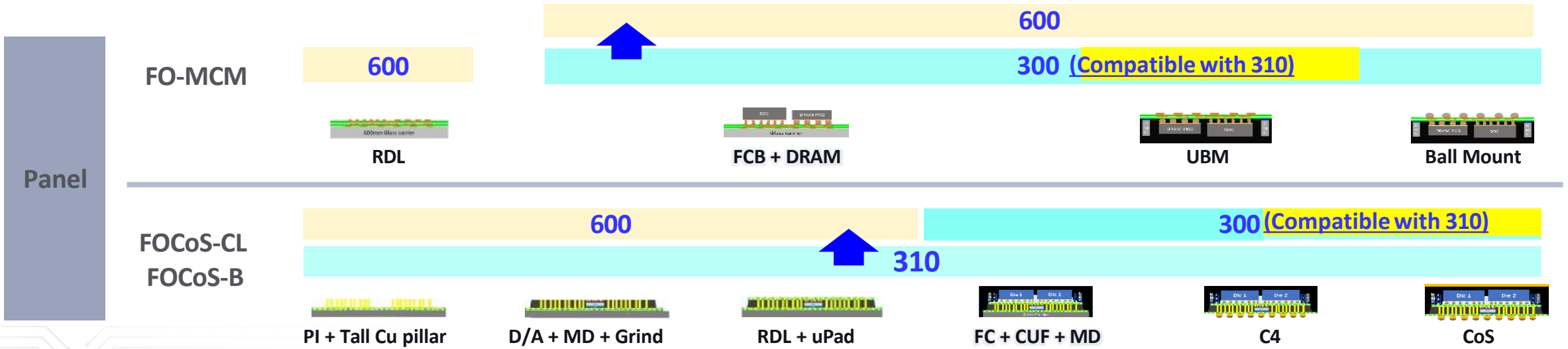
10 Chiplets w 10 silicon bridges



Fanout construction with stack & stagger vias by LDI

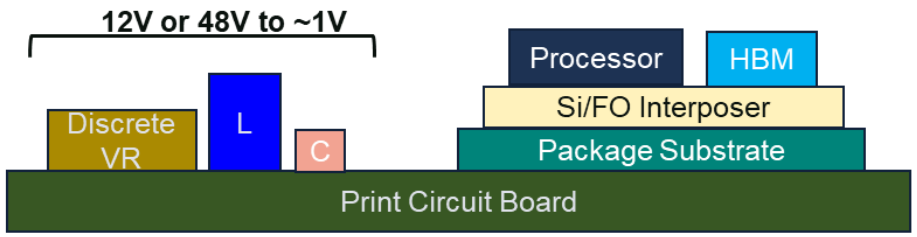
日月光面板級封裝路線圖

Platform	Application	PKG	Remark
 600 Panel	Mobile	FO-MCM	• L/S=5/5 or above • FORDL as a Substrate
 310 Panel	HPC/AI	FOCoS-CL FOCoS-Bridge	• For L/S=2/2 or Less • Bridge and IPD



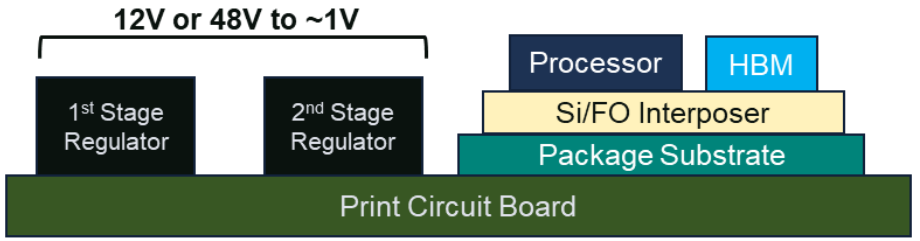
20

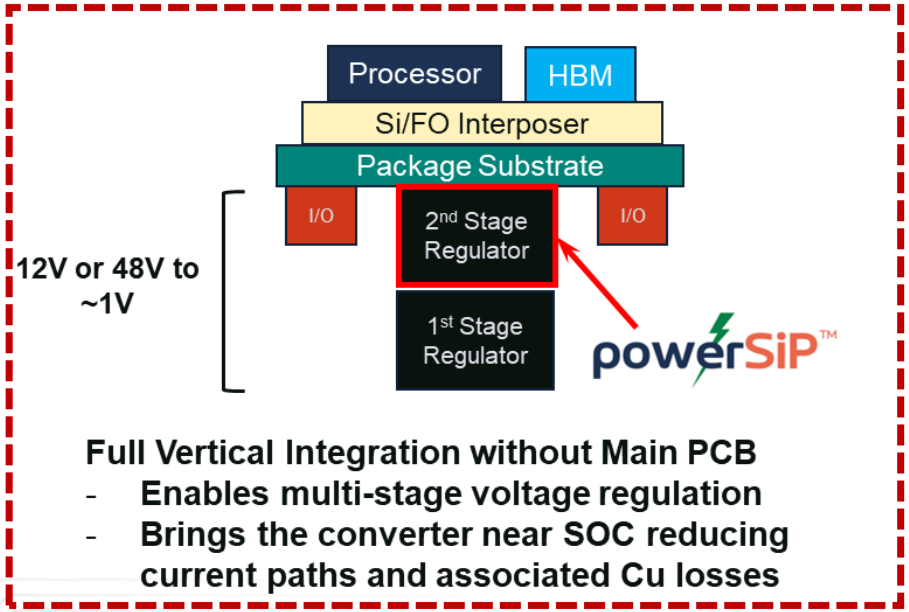
高效能運算系統的電源整合



Typical systems using single stage 12V to ~1V

Side-by-Side Integration

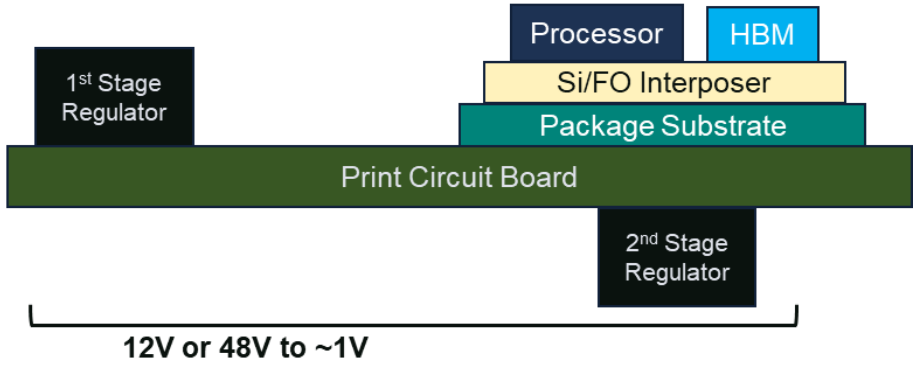




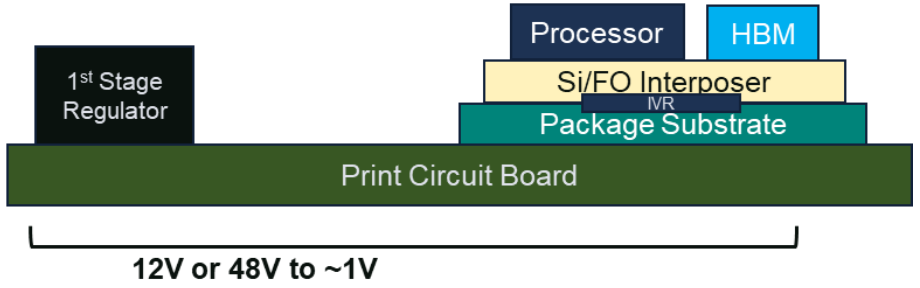
Full Vertical Integration without Main PCB

- **Enables multi-stage voltage regulation**
- **Brings the converter near SOC reducing current paths and associated Cu losses**

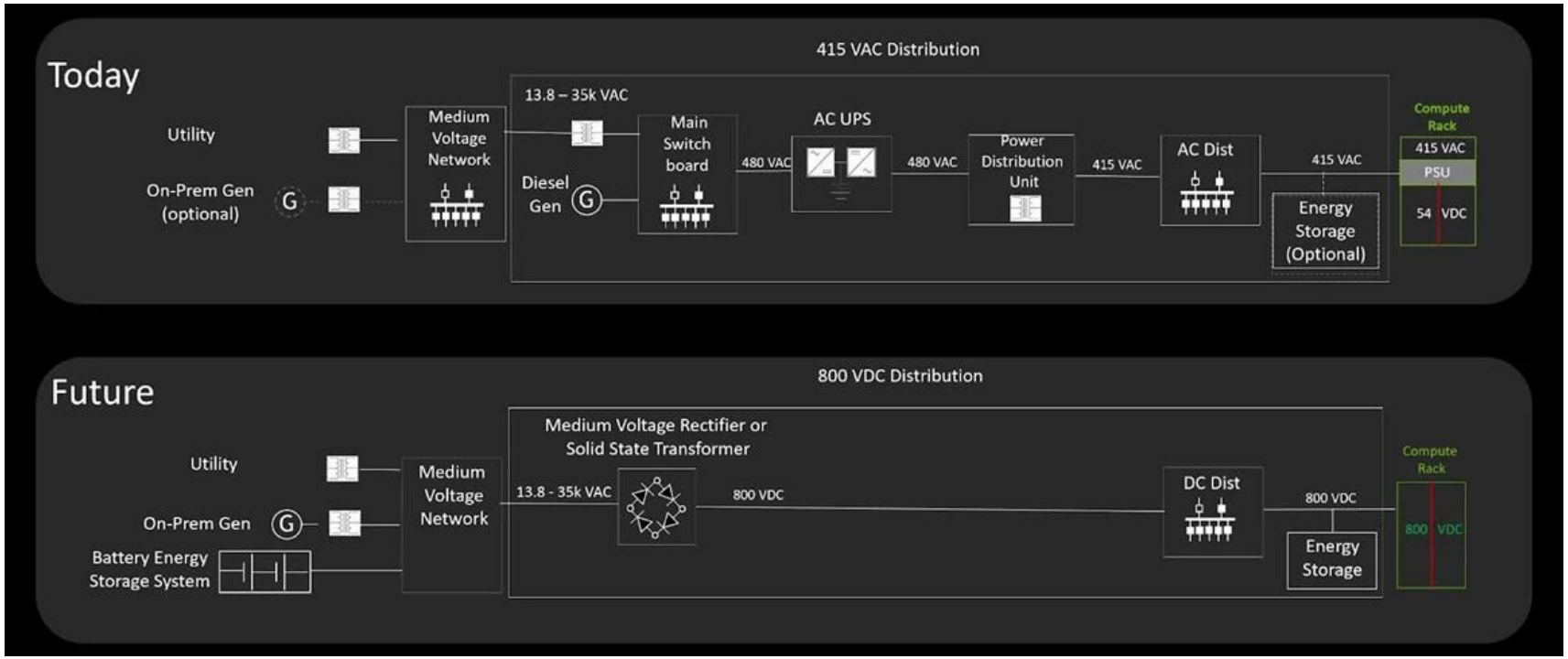
2nd Stage Power with Vertical Integration



2nd Stage Power with IVR Integration



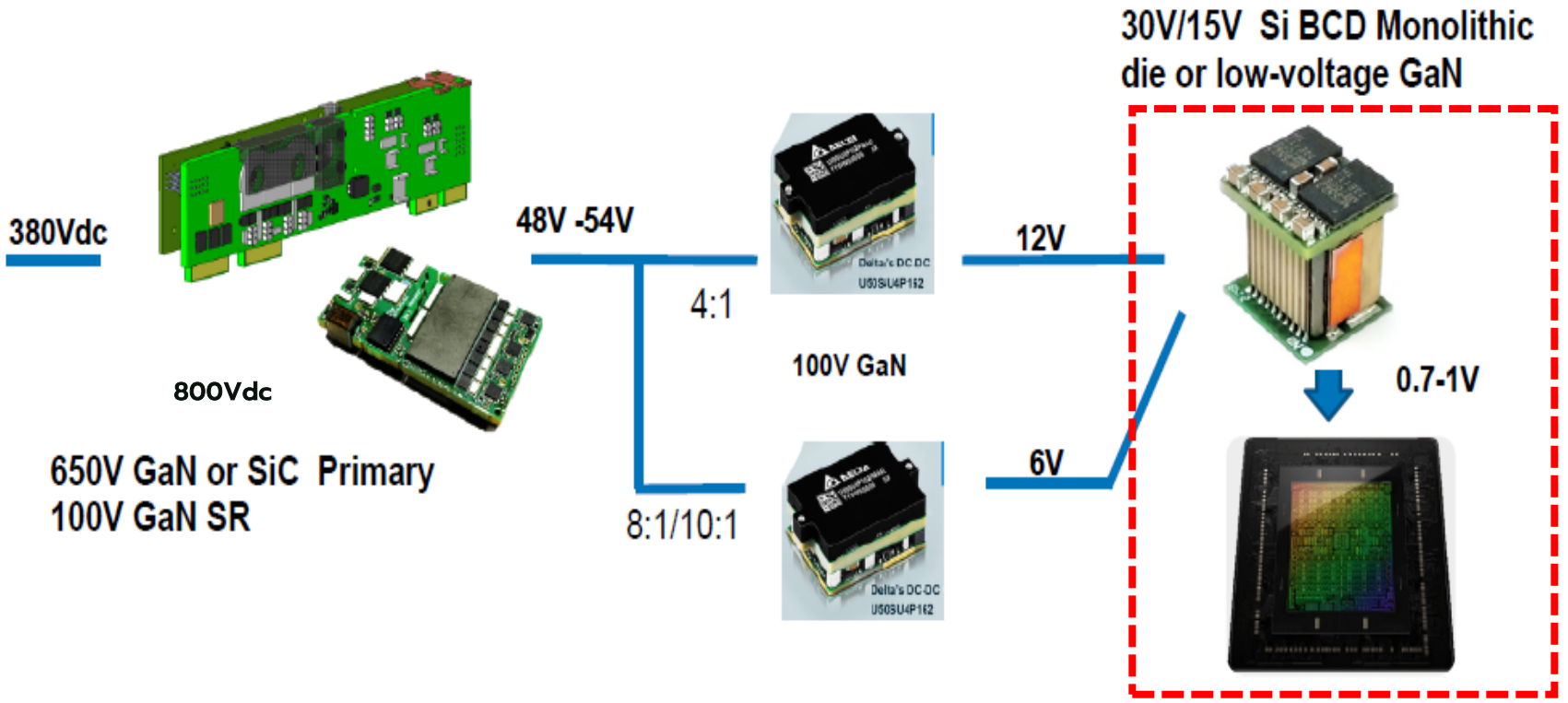
轉換至800V 高壓直流輸電系統



Ref: Nvidia

- Transitioning to 800 VDC power distribution coupled energy storage integration
- Simpler DC distribution system with fewer points of failure, better overall system reliability
- Aligns with EV and renewable industry

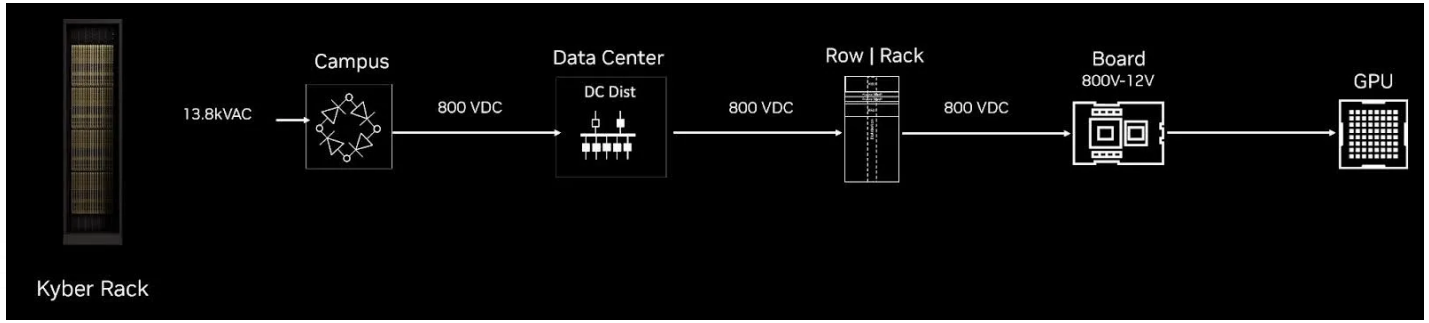
高壓直流電至GPU 供電電壓



GaN devices have superior performance (higher at 100V, 650V); cost still two times higher than Si

BCD Bipolar-CMOS-DMOS

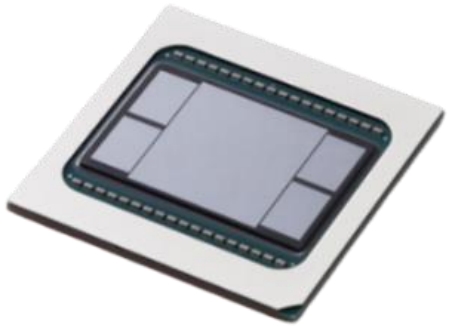
Cyntec, APEC 2025



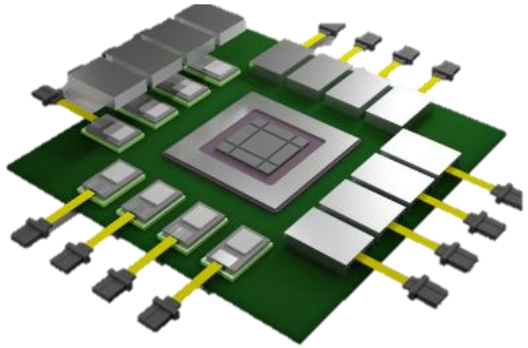
Ref: Nvidia

異質整合：矽光子

Chiplets



CPO
(Co-Packaged Optics)



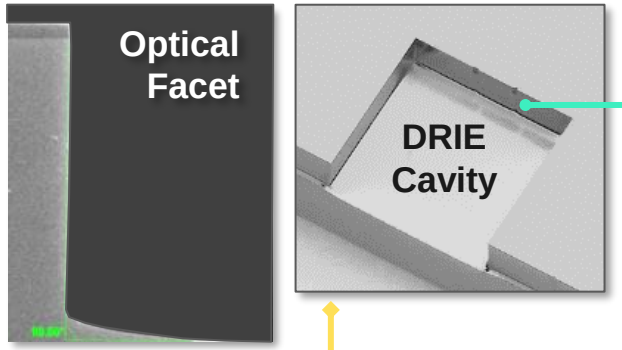
Data Center



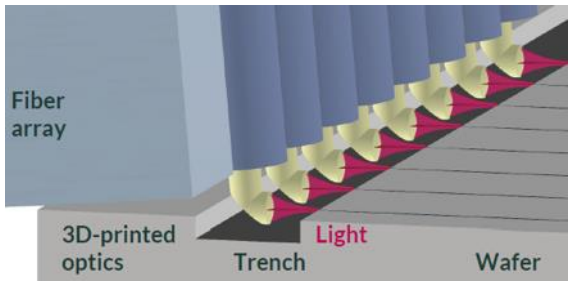


用於光頻寬的SiPh 光引擎 (OE) 工具箱

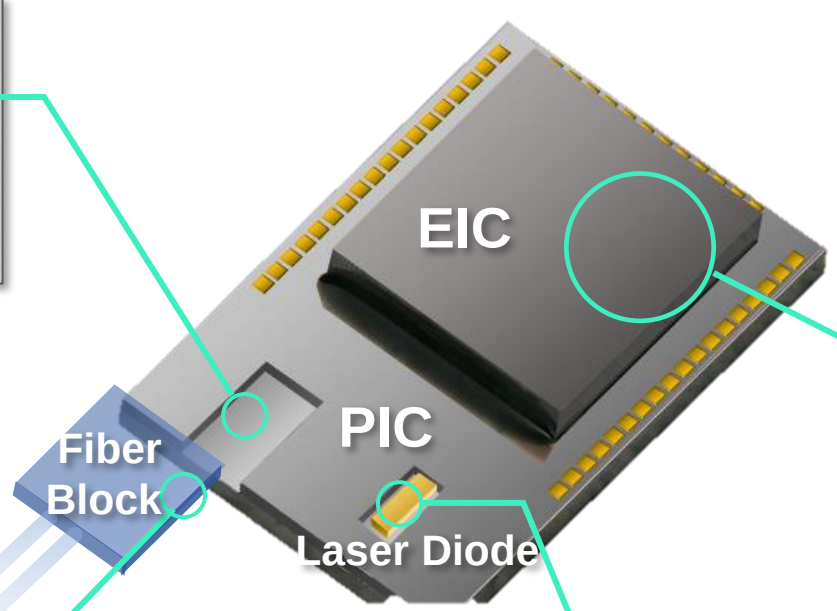
Optical Coupling



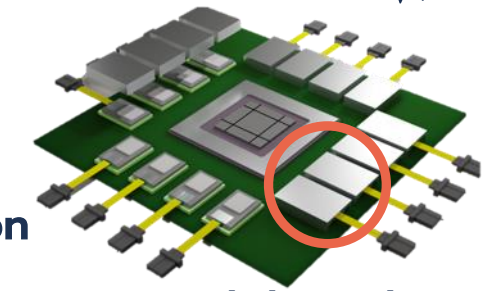
Enable 5x UPH



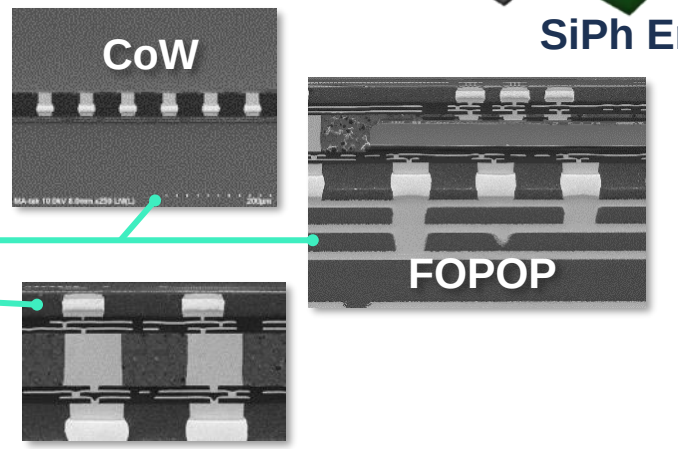
Fiber Passive Alignment

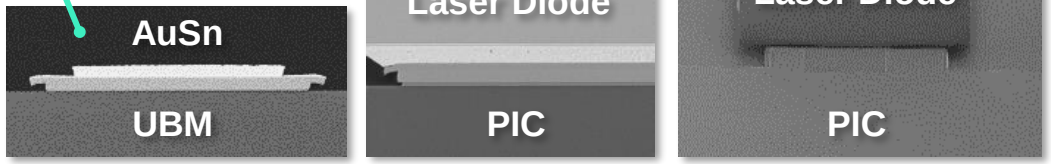


3D EIC/PIC Integration



SiPh Engine



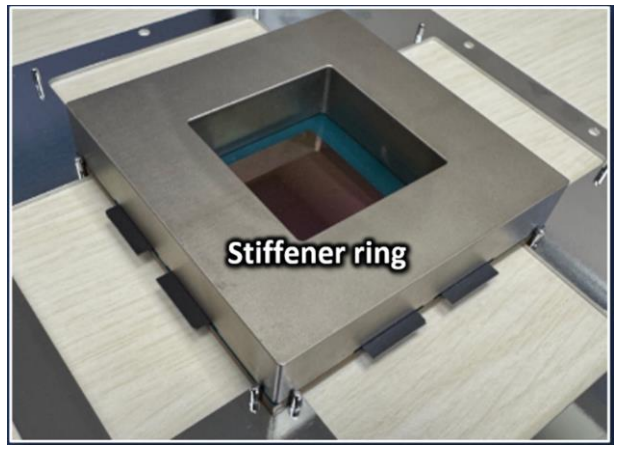


Laser Diode Integration ⟷ Sub-μm Accuracy

OE 與ASIC on Substrate 的挑戰

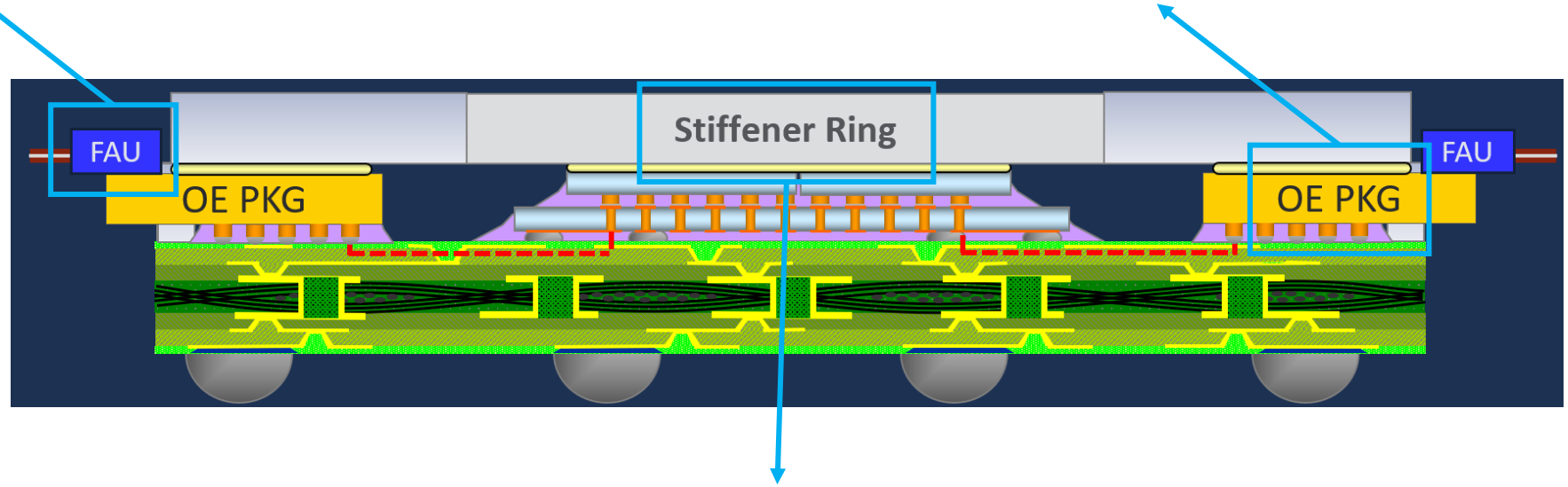
Challenge 1: Fiber Attach

- Coupling approach (PA/AA)
- FAU Detachability and reflow compatibility



Challenge 2: Optical Engine

- Engine size & EIC/PIC integration approach & warpage control
- OE test solution (KGOE) & ensure optical facet quality

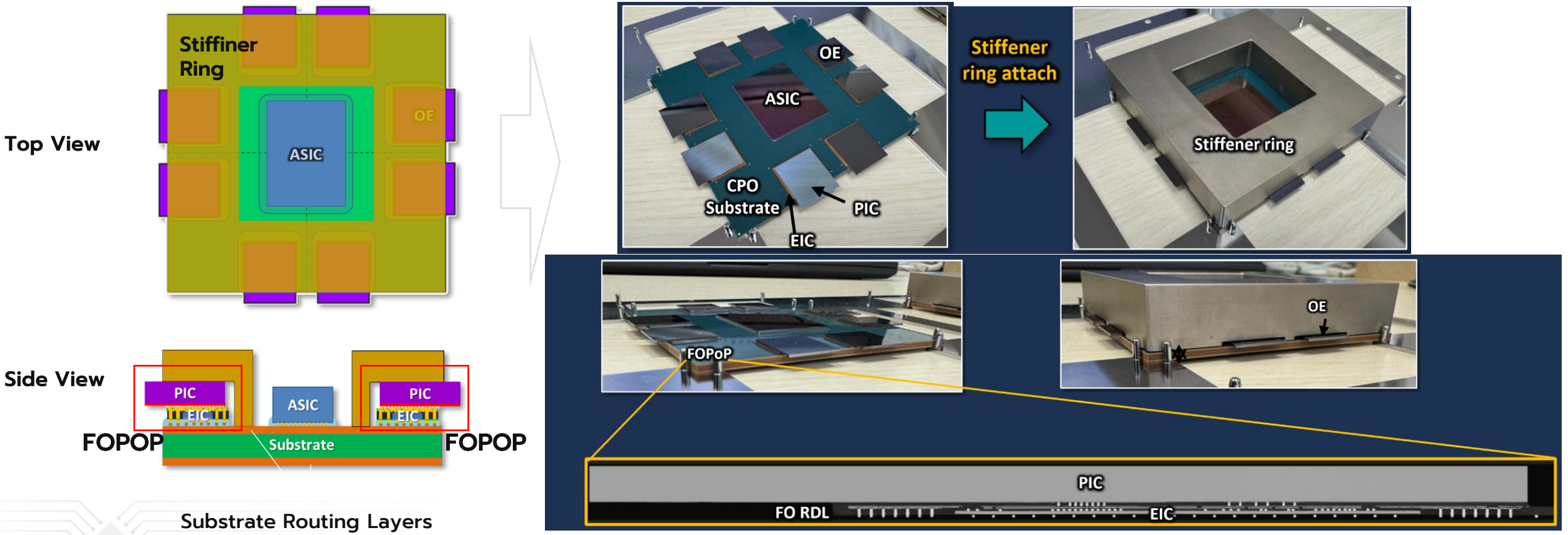


Challenge 3: On Substrate

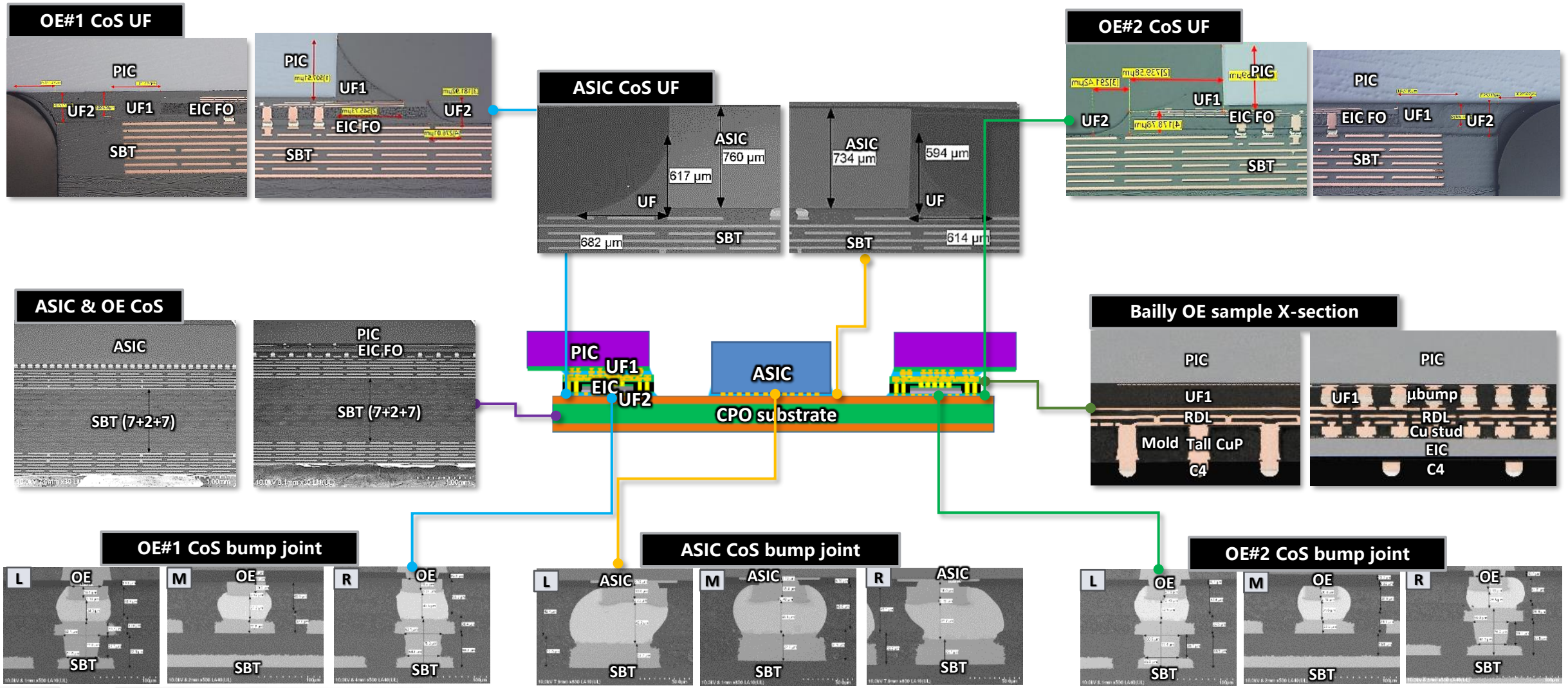
- Overall warpage control
- Thermal dissipation

CPO 測試載具演示（OE on Substrate 製程）

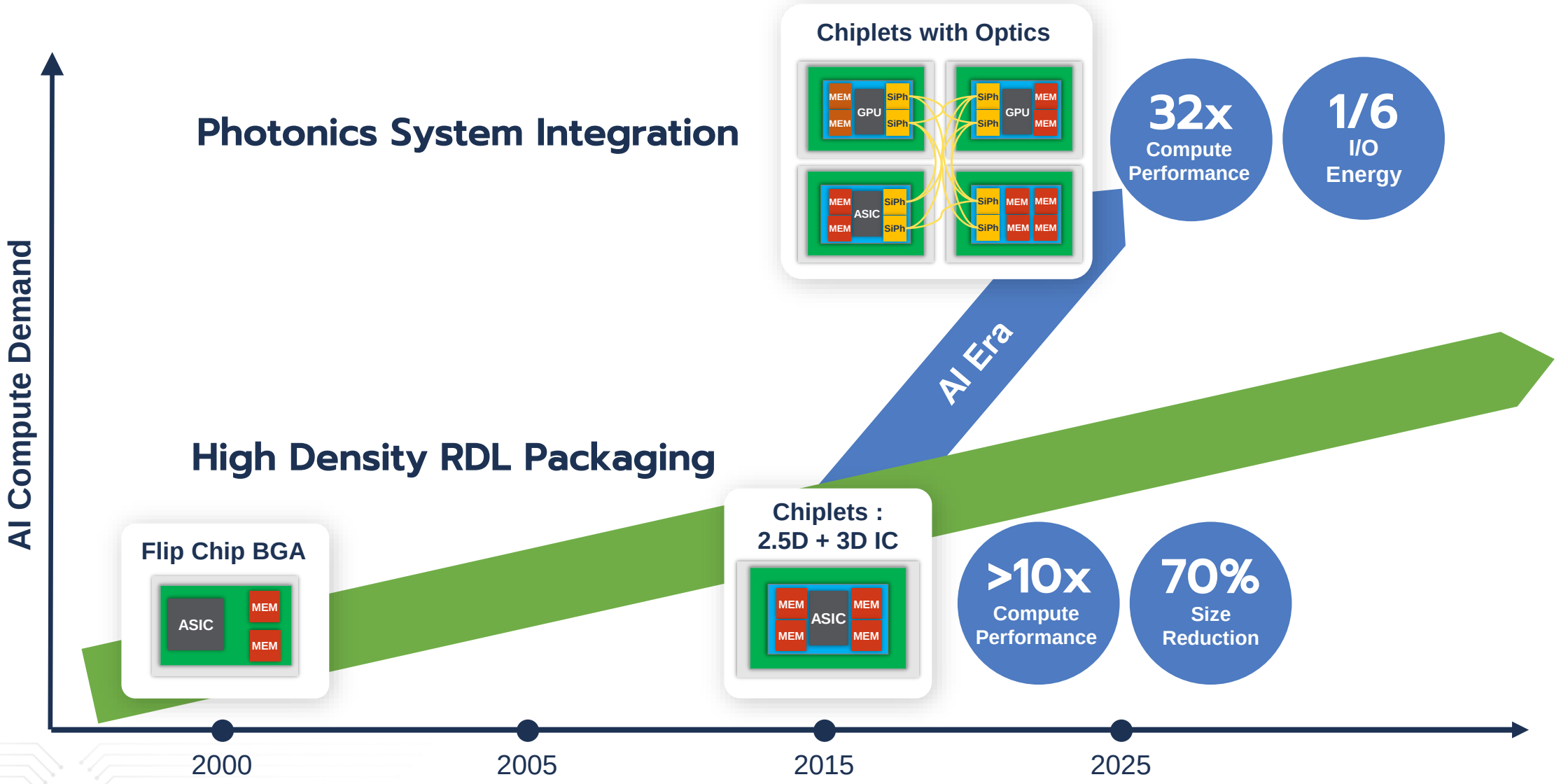
- CPO substrate & stiffener ring & ASIC/EIC designed and manufactured by ASE
- ASIC & 8x OE FCB on substrate with one time reflow process; **>75 mm x 75 mm package size**
- Good bump joint by X-ray and cross section check



共同封裝光學解決方案

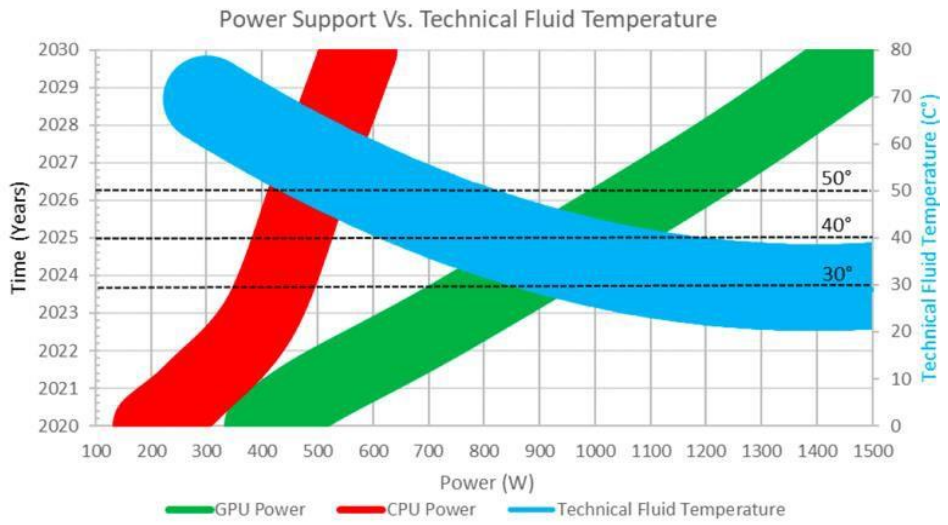

OE#1 CoS UF
PIC
UF2
UF1
EIC FO
SBT
PIC
UF1
EIC FO
UF2
SBT
ASIC CoS UF
ASIC
760 µm
617 µm
UF
682 µm
SBT
ASIC
734 µm
594 µm
UF
SBT
614 µm
OE#2 CoS UF
PIC
UF1
UF2
EIC FO
SBT
PIC
EIC FO
UF1
UF2
SBT
ASIC & OE CoS
ASIC
SBT (7+2+7)
PIC
EIC FO
SBT (7+2+7)
PIC
UF1
EIC
UF2
ASIC
CPO substrate
Bailly OE sample X-section
PIC
UF1
RDL
Mold
Tall CuP
C4
PIC
UF1
µbump
RDL
Cu stud
EIC
C4
OE#1 CoS bump joint
L
OE
SBT
M
OE
SBT
R
OE
SBT
ASIC CoS bump joint
L
ASIC
SBT
M
ASIC
SBT
R
ASIC
SBT
OE#2 CoS bump joint
L
OE
SBT
M
OE
SBT
R
OE
SBT

AI 系統的封裝演化


Chiplets with Optics
Photonics System Integration
MEM
GPU
SiPh
ASIC
32x
Compute
Performance
1/6
I/O
Energy
AI Era
High Density RDL Packaging
Flip Chip BGA
Chiplets :
2.5D + 3D IC
>10x
Compute
Performance
70%
Size
Reduction
AI Compute Demand
2000
2005
2015
2025

AI 先進散熱解決方案

Rising Power Demands Drive Next-Gen Cooling Requirements

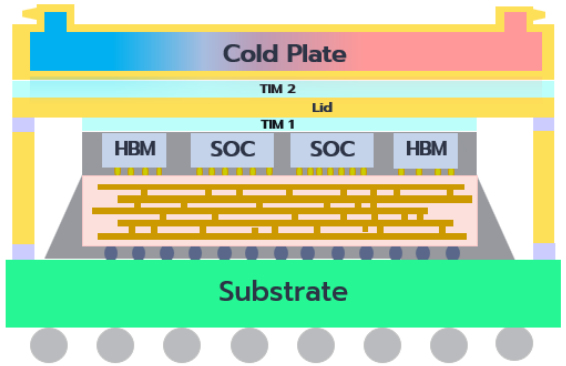


Source:AMD

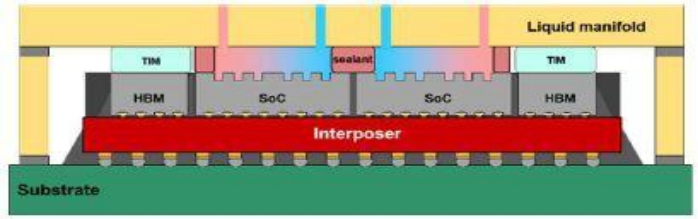
- AI/compute power rapidly increasing:
  - **GPU power > 1500W**
  - **CPU power > 600W (expected by 2028)**
- Thermal fluid temperature must decrease to effectively dissipate increasing heat
- **Thermal management is now a critical bottleneck** for enabling efficient liquid cooling and maintaining lower operational temperatures

Current vs. Next-Gen Package Level Thermal Solutions

System Solution



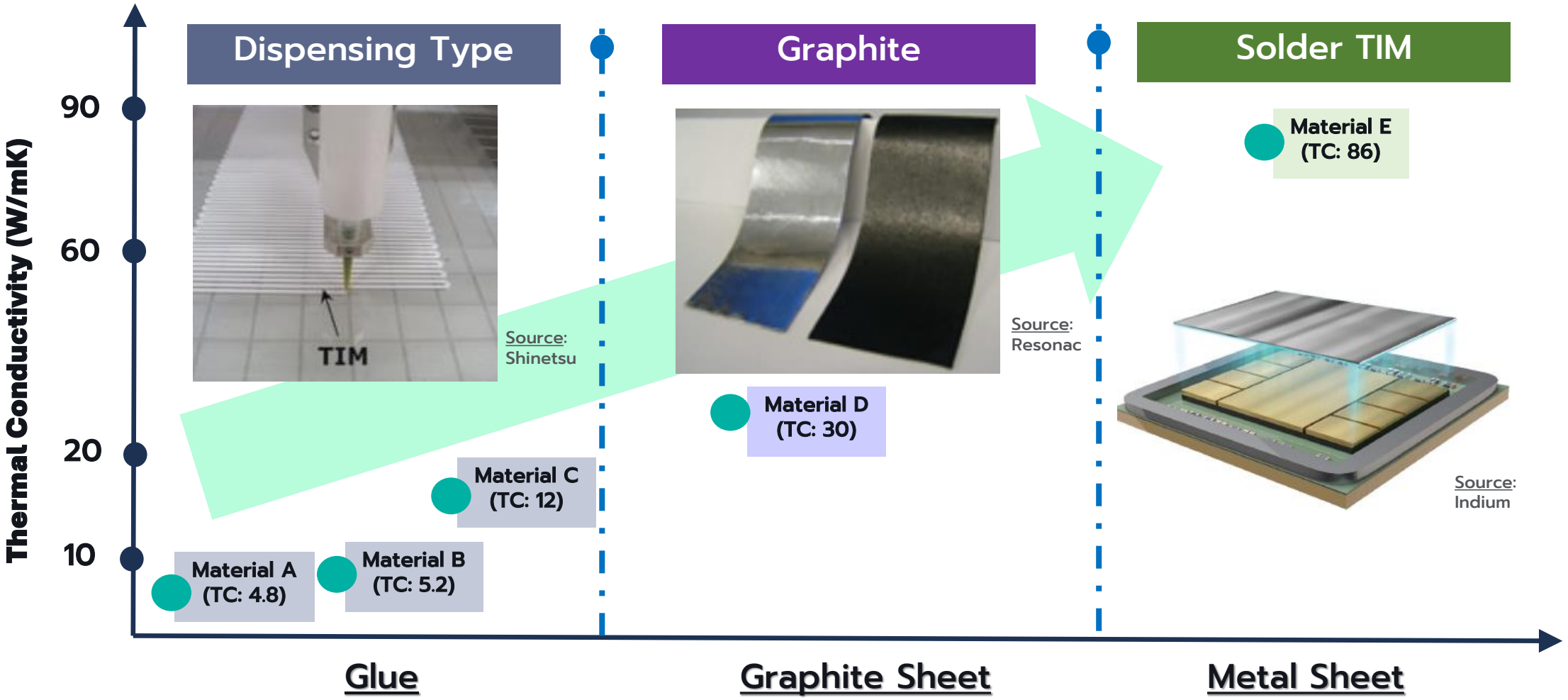
Silicon Solution



TSMC

- **Current:** Uses Thermal Interface Material (TIM) between die/lid and cold plate, but interface resistance degrades performance
- **Next:** Direct liquid-cooled solutions eliminate thermal interfaces, improving dissipation and supporting higher chip power densities

日月光散熱材料解決方案



創新先進封裝迎接AI 挑戰

= Demand = Solution

Packaging Architecture & Technology

Performance	Memory	Area
2x – 7x increased compute workload	2x - 10x network bandwidth & memory capacity	2x - 12x reticle size 2x-6x PKG size (~120x180 mm²)

- ✓ Chiplets & Heterogeneous Integration (CPU, GPU, APU, XPU, HBM, AI accelerator, IPD, DTC)
- ✓ 2.5D Si TSV, FOCoS-CF/CL, FOCoS-B, SiPh, Panel FOCoS

Packaging Infrastructure & Ecosystem

Power	Thermal
Exascale 21 MW to zettascale 500 MW power delivery	~3 kW thermal coolings

Power	Thermal
✓ PowerSiP (VRM, IVR Integration) ✓ Backside power ✓ CPO	✓ Cooling structure, liquid cooling, Si microchannel, new TIM materials

ASE Confidential Proprietary

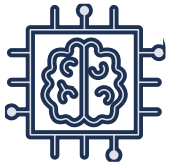
總結



- **AI and Data** continue to fuel semiconductor innovation, with exponential proliferation through 2030 and beyond that **will shape global life and lifestyle in unimaginable ways.**



- Collectively, our industry is accelerating the AI economy through **Heterogeneous Integration advancements.**



- Packaging creativity is enabling seamless integration of multiple chiplets, SiPs, modules into one single package optimized for **enhanced functionality and operating characteristics.**

Thank you